Provably Learning Diffusion Models under the Manifold Hypothesis: Collapse and Refine

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Abstract

Diffusion models generate high-dimensional data with remarkable quality, yet how their training efficiently learns the score function, bypassing the curse of dimensionality when data is supported on low-dimensional manifolds, remains theoretically unexplained. We identify a collapse-and-refine mechanism driven by the geometry of the score function itself: at small noise scales, the diverging singularity of the score drives a rapid dimensional collapse of the induced denoising map onto the data manifold projection; at moderate noise scales, training refines the intrinsic density on the learned manifold. We instantiate this principle as Score-induced Latent Diffusion (SiLD), a two-stage framework in which both manifold learning and density estimation emerge from a single denoising score matching objective, replacing the heuristic KL regularization of VAE-based latent diffusion models. We prove that the resulting sample complexity depends on the intrinsic dimension rather than the ambient dimension. Experiments on Stacked MNIST, CelebA variants, and molecular generation benchmarks show that SiLD matches or outperforms VAE-based LDMs in generation quality and consistently improves reconstruction, validating our theoretical predictions.

1 Introduction

Diffusion models have emerged as a dominant paradigm for generative modeling, demonstrating remarkable capability in synthesizing high-fidelity samples from complex, high-dimensional data distributions [50, 27, 51, 52]. The connection between these models and score matching, particularly through the lens of denoising autoencoders, has been firmly established [57, 27]. Despite their empirical success, the theoretical foundations enabling efficient learning in high-dimensional spaces remain a subject of intense inquiry. A central puzzle lies in the "curse of dimensionality": theoretically, learning a probability distribution in an ambient space of dimension d typically requires a sample

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complexity exponential in d [58, 5]. The prevailing resolution to this paradox is the manifold hypothesis, which posits that real-world data, while embedded in a high-dimensional ambient space, (approximately) resides on a low-dimensional manifold of intrinsic dimension $k \ll d$ [18, 37].

Recent theoretical works have leveraged this low-dimensional structure to establish improved bounds on statistical estimation and sampling complexity [15, 10, 41, 34, 55, 44, 2], proving that sample complexity depends on the intrinsic dimension k rather than d when the score is well-approximated. Notably, [33] and [28] showed that the DDPM sampler automatically adapts to unknown low-dimensional structure, achieving iteration complexity scaling nearly linearly in k without any prior knowledge of the manifold. From a structural perspective, [42] and [53] proved that trained diffusion models detect and encode the data manifold by approximating its normal bundle, while [16] showed that score smoothing implicitly regularizes toward manifold-adaptive solutions. More recently, [6, 20, 32] investigated how diffusion and flow matching models adapt to low-dimensional structures.

However, these analyses predominantly focus on the properties of the converged score estimator, treating the optimization process abstractly or relying on specific architectural assumptions such as single-layer networks [6]. While recent works have begun to probe the training dynamics of diffusion models, [49] provided the first provably efficient result linking gradient descent on the DDPM objective to recovering mixture model parameters, and [61] proved that optimizing the diffusion training loss with a low-rank parameterization is equivalent to a subspace clustering problem; yet these results are confined to restricted model classes and do not characterize the fine-grained weight evolution of general deep networks. From the broader neural network optimization perspective, mean-field theory [38, 11, 54] provides a powerful framework for analyzing two-layer network training dynamics, and feature learning theory [14, 39, 1] has shown that gradient-based training discovers low-dimensional relevant subspaces through saddle-to-saddle dynamics. Yet how these mechanisms manifest specifically in the score matching setting, and whether the optimization process itself exploits the low-dimensional geometric structure of the data, remains largely unexplored. This raises a fundamental question that current theory cannot answer:

How does a neural network, initialized isotropically, adaptively discover the low-dimensional data support and efficiently learn the intrinsic distribution density amidst high-dimensional noise?

We answer this question by proposing *Score-induced Latent Diffusion (SiLD)*, a theoretically grounded framework that characterizes the gradient descent dynamics of score matching on low-dimensional manifolds. The key insight is that the singularity of the score function at small noise levels naturally induces a *two-stage* learning mechanism: the network first discovers the geometry of the data manifold, and then refines the intrinsic probability density within it. This key insight naturally induces a novel two-stage training strategy: first learning the manifold with low-noise score matching and then learning the density on manifold. Both stages are trained under a single DDPM objective, without any auxiliary losses or heuristic regularization; the latent representation is induced by the score function itself rather than imposed by a separate encoder. Our main contributions are as follows:

- **Convergence Guarantees.** We prove quantitative convergence rates for both stages. In Stage 1, a mean-field gradient flow analysis shows that the geometric alignment risk decays exponentially fast. In Stage 2, we establish generalization bounds via Random Feature regression on the low-dimensional manifold, proving that the excess risk depends polynomially on the intrinsic dimension k and the sample size n, independent of the ambient dimension d.

- **End-to-End Sample Complexity.** We establish an end-to-end sampling guarantee. The manifold-regime contribution achieves a Wasserstein-2 rate depending only on the intrinsic dimension. The high-noise contribution, handled by an auxiliary random-feature head, contributes a polynomial-in-d term that is exponentially damped by the integration time. Together, this avoids the curse of ambient dimensionality.

- **Empirical Validation.** We validate our theoretical predictions on Stacked MNIST, CelebA, and molecular generation benchmarks, demonstrating that SiLD matches or outperforms VAE-based latent diffusion models [47] in generation quality, confirming that the score matching objective alone is sufficient to drive both manifold learning and density estimation.

2 Related Work

Statistical Theory of Diffusion on Manifold. Beyond the convergence and adaptivity results discussed in the introduction, several works have examined more fine-grained aspects of diffusion models on manifolds. [3] proved nearly d-linear convergence bounds for diffusion models via stochastic localization, establishing that the number of reverse steps scales nearly linearly in the intrinsic dimension. Most recently, [8] introduced the (p, q)-Wasserstein dimension and proved the first Wasserstein-p convergence guarantee for diffusion models under a finite-moment condition only, without compact support, manifold, or smooth density assumptions, achieving the sharpest known rates to date. [9] showed that even with inexact score estimates, generated samples tend to drift along rather than away from the manifold, while [19] proved that OT-CFM dynamics on manifold-supported targets contracts exponentially in normal directions and remains neutral along tangential directions. [36] identified the score singularity in the normal direction as a hindrance to sampling accuracy and proposed methods to mitigate it. From an analytical perspective, [21] derived asymptotically exact learning curves for denoising score matching with random feature networks on manifold data, confirming that sample complexity scales linearly with the intrinsic dimension for linear manifolds while showing that this benefit diminishes for nonlinear manifolds — a limitation that our two-stage decoupling is designed to address. Our work complements this body of literature by shifting focus from the statistical convergence properties to the optimization dynamics.

Training Dynamics of Generative Models. The theoretical study of how neural networks learn during diffusion training remains substantially less developed than their statistical theory. [24] leveraged the Neural Tangent Kernel (NTK) [29] to establish the first generalization bounds for gradient-descent-trained networks on the score matching objective, though NTK analyses cannot capture the feature learning dynamics that arise in practice. [49] and [61] showed that gradient descent on the DDPM objective recovers low-dimensional structure, respectively linking it to Gaussian mixture recovery and subspace clustering. [23] and [35] further analyzed feature learning and representation dynamics in diffusion models through low-dimensional data models. Complementary lines of work analyze diffusion training via high-dimensional asymptotics and convex optimization [13, 12, 64, 63] or establish coarse-to-fine spectral dynamics in sampling [60, 59]. Recently, [7] identified an implicit dynamical regularization mechanism via two training timescales; our work provides the geometric counterpart, proving that the "collapse-then-refine" mechanism is the structural driver behind why memorization is dynamically postponed and showing that the score singularity drives an analogous geometry-before-density hierarchy.

Latent Diffusion Models. Latent diffusion models (LDMs) [47] achieve state-of-the-art generation quality by operating in a compressed latent space learned by a VAE [30], rather than directly in the high-dimensional pixel space. While highly effective in practice, this approach introduces a fundamental tension: the VAE encoder is trained with a heuristic KL regularization term to encourage a well-structured latent space, which is independent of and potentially misaligned with the score matching objective used in the diffusion stage. Several works have attempted to bridge this gap by jointly training the encoder and diffusion model [56] or by revisiting how the latent itself is trained [25]. Our work, SiLD, offers a principled alternative: both manifold learning and density estimation emerge naturally from the score matching objective at different noise scales, eliminating the need for KL regularization entirely. This provides a theoretical justification for why the latent space induced by the score function is geometrically well-suited for diffusion, and consistently improves reconstruction quality as validated in our experiments.

3 Score-induced Latent Diffusion Model

3.1 Preliminaries

Notation. We use $\|\cdot\|$ to denote the Euclidean norm for vectors and the Frobenius norm for matrices unless otherwise specified. For a matrix A, $\|A\|_{\mathrm{op}}$ denotes its operator norm. We employ standard asymptotic notations such as $O(\cdot)$, $\Omega(\cdot)$, and $o(\cdot)$. We write $f \asymp g$ if $f = O(g)$ and $g = O(f)$.

Data distribution. Let p_{data} denote the unknown target distribution supported on a k-dimensional compact smooth manifold $\mathcal{M}$ embedded in $\mathbb{R}^d$, with $k \ll d$. For any point x in the tubular neighborhood of $\mathcal{M}$, we denote by $\Pi_{\mathcal{M}}(x)$ its projection onto $\mathcal{M}$ and by $d_{\mathcal{M}}(x) = \inf_{z\in\mathcal{M}} \|x - z\|$ its distance to $\mathcal{M}$. The tangent and normal spaces of $\mathcal{M}$ at a point z are denoted $T_z\mathcal{M}$ and $N_z\mathcal{M}$,

respectively. We denote by $\tau > 0$ the reach of $\mathcal{M}$, i.e., the largest radius such that the projection $\Pi_{\mathcal{M}}$ is uniquely defined within the tubular neighborhood $U_\tau = \{x \in \mathbb{R}^d : d_{\mathcal{M}}(x) < \tau\}$.

Diffusion process. We consider the Variance-Preserving (VP) forward process [27], which corrupts a clean sample $x_0 \sim p_{\text{data}}$ as $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_d)$, where $\{\bar{\alpha}_t\}_{t\in[0,T]}$ is a monotonically decreasing noise schedule. We denote the marginal distribution of x_t by p_t, and write $h(t) := 1 - \bar{\alpha}_t$ for the noise variance at time t. The score function of p_t is $s^*(x,t) := \nabla_x \log p_t(x)$.

Score matching objective. Following [27, 57], the score function is estimated by minimizing the denoising score matching (DSM) objective: $\mathcal{L}(\theta) = \mathbb{E}_{t,x_0,\varepsilon}\left[\left\| s_\theta(x_t,t) + \varepsilon/\sqrt{h(t)} \right\|^2\right]$, where the expectation is over $t \sim \text{Unif}[0,T]$, $x_0 \sim p_{\text{data}}$, and $\varepsilon \sim \mathcal{N}(0, I_d)$. It is well established [57] that minimizing $\mathcal{L}(\theta)$ is equivalent to minimizing the explicit score matching error $\mathbb{E}[\|s_\theta(x_t,t) - s^*(x_t,t)\|^2]$ up to a constant.

3.2 Manifold Hypothesis and Score Singularity

Manifold hypothesis. We operate under the manifold hypothesis, which posits that p_{data} is supported on a k-dimensional compact smooth manifold $\mathcal{M} \subset \mathbb{R}^d$ with $k \ll d$. For any point x in the tubular neighborhood U_τ, the perturbed distribution p_t admits the following decomposition of its score.

Proposition 3.1 (Score decomposition [15, 34])**.** *Assume $p_0 \in C^2(\mathcal{M})$ with $0 < p_{\min} \le p_0 \le p_{\max} < \infty$. For $x \in U_{\tau/2}$ and $h(t) \le c\,\tau^2$ with a sufficiently small universal constant $c \in (0,1)$, the score function $s^*(x,t) := \nabla_x \log p_t(x)$ admits the scale-separated decomposition*

$$s^*(x,t) = \underbrace{-\frac{x - \Pi_{\mathcal{M}}(x)}{h(t)}}_{\text{(I): } O(h(t)^{-1})} + \underbrace{\nabla_x \log p_0\big(\Pi_{\mathcal{M}}(x)\big) + \nabla_x H\big(\Pi_{\mathcal{M}}(x),\, x - \Pi_{\mathcal{M}}(x)\big)}_{\text{(II): } O(1)} + O(h(t)), \quad (1)$$

where $H : \{(z,\nu) : z \in \mathcal{M},\, \nu \in N_z\mathcal{M}\} \to \mathbb{R}$ is a smooth function.

Term *(I)* is the *normal restoring force*, pointing from x toward its projection on $\mathcal{M}$. Term *(II)* is the *intrinsic-density and geometric correction*: the first summand $\nabla_x \log p_0(\Pi_{\mathcal{M}}(x))$ denotes the Euclidean gradient of the composition $x \mapsto \log p_0(\Pi_{\mathcal{M}}(x))$, which lies in $T_{\Pi_{\mathcal{M}}(x)}\mathcal{M}$ and encodes the intrinsic density gradient; the second summand accounts for the geometry of the embedding. Explicitly, H combines two contributions, $H(z,\nu) = -\frac{1}{2}\langle \nu, z\rangle - \frac{1}{2}\log\det\big(I_k - B_\nu\big)$, where the first term is a *VP-shrinkage correction* arising from the rescaling $y = x/\sqrt{1-h(t)}$, and the second is an *extrinsic-curvature correction* with $(B_\nu)_{ij} := \langle \nu, \mathrm{II}(e_i, e_j)\rangle$ and II the second fundamental form of $\mathcal{M}$.

Implications of scale separation. Proposition 3.1 reveals a fundamental asymmetry in the learning problem. As $h(t) \to 0$, the normal restoring force dominates the score by a factor of $O(h(t)^{-1})$, creating a strong geometric signal that dwarfs the tangential density information. This *scale separation* [34] has three consequences. First, learning the geometry of $\mathcal{M}$ is statistically easier than learning the intrinsic density p_0: at small noise scales, the DSM loss is dominated by the projection term and provides a strong gradient signal toward approximating $\Pi_{\mathcal{M}}(\cdot)$ before the residual density term becomes relevant. Second, the dominant term *(I)* is the gradient of a scalar potential, $-\frac{x-\Pi_{\mathcal{M}}(x)}{h(t)} = -\nabla_x\left(\frac{d_{\mathcal{M}}(x)^2}{2\,h(t)}\right)$, by Federer's identity $\nabla_x\big(\frac{1}{2}\,d^2_{\mathcal{M}}(x)\big) = x - \Pi_{\mathcal{M}}(x)$. This conservative structure motivates a conservative-form architecture (Section 3.3), whose output is structurally constrained to conservative vector fields and therefore aligned with the geometric category of the target score at small noise.

The optimization challenge. The same singularity that enables efficient geometry learning poses an obstacle for simultaneous density learning: at small noise scales, the $O(h(t)^{-1})$ magnitude of the normal component dominates the DSM loss, drowning out the $O(1)$ density signal [36]. A single network trained end-to-end on the DSM objective must therefore resolve two tasks operating at incompatible scales, leading to inefficient optimization. This motivates our *Score-induced Latent Diffusion (SiLD)* framework, introduced in Section 3.3, which decouples the two stages by exploiting the scale separation structure of Proposition 3.1.

Algorithm 1 Score-induced Latent Diffusion (SiLD) Training

Require: Dataset $\mathcal{D} = \{x_i\}_{i=1}^n \subset \mathbb{R}^d$, noise levels $h(t_1) \ll h(t_2)$, regularization λ

Stage 1 (Geometric Alignment, at noise level $h(t_1)$):

Train $f_1 \in \mathcal{F}_1$ via gradient descent on the DSM loss to learn the manifold projection

Compute $\hat{x} := x + h(t_1) \cdot f_1(x) \approx \Pi_{\mathcal{M}}(x)$

Stage 2 (Density Estimation, at noise level $h(t_2)$):

Train $f_2 \in \mathcal{F}_2$ on the residual score: $\hat{f}_2 = \arg\min_{f_2 \in \mathcal{F}_2} \mathbb{E}_{t,x} \| f_2(\hat{x}, t) - s^*_{\text{res}}(x, t) \|^2 + \lambda R(f_2)$

return Score network $s_\theta(x, t) = -\dfrac{x - \hat{x}}{h(t)} + \hat{f}_2(\hat{x}, t)$

3.3 Score-induced Latent Diffusion: A Two-Stage Framework

The scale separation in Proposition 3.1 motivates a principled decoupling of the score learning problem into two sequential stages, each targeting one term in the decomposition (1). We call this framework *Score-induced Latent Diffusion (SiLD)* (with generic algorithm presented in Algorithm 1), reflecting that the latent representation is induced by the geometric structure of the score function itself, rather than imposed by an auxiliary objective. For tractability of the analysis, Stage 1 uses a two-layer conservative-form network whose gradient-of-potential structure matches the conservative target score at small noise, and Stage 2 a random-feature network operating on the Stage 1 projection.

Stage 1: Geometric Alignment. At small noise scales $h(t_1) \ll 1$, the DSM loss is dominated by the normal restoring force *(I)*. Since this dominant term is a conservative vector field, we constrain our network to the same class by parametrizing it as the gradient of a scalar potential. Concretely, we train a *conservative-form* two-layer network:

$$f_1(x; \theta) \;=\; \frac{1}{h(t_1)} \Big(W \,\mathrm{diag}(a)\, \sigma(W^\top x + b) \;-\; x \Big), \tag{2}$$

with parameters $\theta = (W, a, b)$, where $W = [w_1, \ldots, w_m] \in \mathbb{R}^{d \times m}$, $a \in \mathbb{R}^m$, $b \in \mathbb{R}^m$, and $\sigma(\cdot)$ is a nonlinear activation. Per-neuron, $f_1(x; \theta) = \frac{1}{h(t_1)} \big(\sum_{j=1}^m a_j \, w_j \, \sigma(w_j^\top x + b_j) - x \big)$: each neuron contributes a vector along its input direction w_j, scaled by $a_j \sigma(w_j^\top x + b_j)$. This direction-parallel structure is precisely what makes f_1 a conservative vector field: as we show in Lemma 4.1, $f_1 = -\nabla_x \Phi_{\text{net}}$ for an explicit scalar potential Φ_{net}, matching the category of the target score at small noise. Minimizing the DSM objective drives the induced projection map to align with $\Pi_{\mathcal{M}}(x)$:

$$\hat{x} \;:=\; h(t_1)\, f_1(x; \theta) + x \;=\; W \,\mathrm{diag}(a)\, \sigma(W^\top x + b) \;\approx\; \Pi_{\mathcal{M}}(x). \tag{3}$$

Stage 2: Density Estimation. Once Stage 1 has converged and W is frozen, we decouple the singular normal component by constructing the full score network as:

$$s_\theta(x, t_2) = -\frac{x - \hat{x}}{h(t_2)} + f_2(\hat{x}, t_2; \theta_2), \tag{4}$$

where $f_2(\hat{x}, t_2; \theta_2) = U \Phi(\hat{x}, t_2)$ is a Random Feature network with frozen features, and $U \in \mathbb{R}^{d \times m}$ is the only trainable parameter. By construction, the singular normal components cancel up to the Stage-1 alignment error, and Stage 2 reduces to estimating the $O(1)$ residual score:

$$\min_{\theta_2} \mathbb{E}_{t_2, x \sim p_{t_2}} \left\| f_2(\hat{x}, t_2; \theta_2) - s^*_{\text{res}}(x, t_2) \right\|^2, \tag{5}$$

where, for $x \in U_{\tau/2}$, $s^*_{\text{res}}(x, t) := s^*(x, t) + \frac{x - \Pi_{\mathcal{M}}(x)}{h(t)}$ is the residual score; by Proposition 3.1, $s^*_{\text{res}}(x, t) = \nabla_x \log p_0\big(\Pi_{\mathcal{M}}(x)\big) + \nabla_x H\big(\Pi_{\mathcal{M}}(x),\, x - \Pi_{\mathcal{M}}(x)\big) + O\big(h(t)\big)$ uniformly over $x \in U_{\tau/2}$.

4 Theoretical Analysis

We now provide rigorous theoretical guarantees for the two stages of SiLD. All proofs are deferred to the Appendix B.

4.1 Stage 1: Dimensional Collapse

We first justify the architectural choice of (2). The key observation is that the dominant term in the score decomposition (1) is a conservative vector field, and our conservative-form network is structurally constrained to the same functional class.

Lemma 4.1 (Structural Alignment)**.** *Let $\sigma \in C^2(\mathbb{R})$ be non-polynomial with primitive $\Psi(z) := \int_0^z \sigma(u)\,du$. Consider the conservative-form network* (2) *with parameters $\theta = (W, a, b)$, and the neural potential*

$$\Phi_{\text{net}}(x;\theta) \;:=\; \frac{\|x\|^2}{2\,h(t)} \;-\; \frac{1}{h(t)} \sum_{j=1}^{m} a_j\, \Psi(w_j^\top x + b_j).$$

Then $f_1(\,\cdot\,;\theta) = -\nabla_x \Phi_{\text{net}}(\,\cdot\,;\theta)$ on $\mathbb{R}^d$ for every θ. Moreover, for any $\varepsilon > 0$, there exist $m = m(\varepsilon, \mathcal{M}, \tau, \sigma) \in \mathbb{N}$ and parameters $\theta = (W, a, b)$ with $W \in \mathbb{R}^{d\times m}$ and $a, b \in \mathbb{R}^m$ such that

$$\sup_{x \in \overline{U_{\tau/2}}} \left\| W\,\mathrm{diag}(a)\,\sigma(W^\top x + b) \;-\; \Pi_{\mathcal{M}}(x) \right\| \;<\; \varepsilon. \tag{6}$$

We then analyze the gradient flow dynamics of Stage 1 under the mean-field framework [38, 11, 54], which captures feature learning and enjoys favorable convergence guarantees. For tractability of the analysis, we adopt a random-feature-style simplification standard in mean-field studies of two-layer networks [38, 11]: the output coefficients a and biases b are frozen at their initial values, and only the input weights w evolve under gradient flow. Lemma 4.1 establishes expressivity of the full conservative-form class; for the frozen-(a, b) dynamics analyzed below, the required approximation and reachability properties are captured by the non-degeneracy condition in Assumption 4.2.

Let $q \in \mathcal{P}_2(\mathbb{R}^{d+2})$ denote the joint distribution over neuron parameters $\theta = (w, a, b) \in \mathbb{R}^d \times \mathbb{R} \times \mathbb{R}$, initialized at $q_0 = \nu_w \otimes \nu_a \otimes \nu_b$, where $\nu_w = \mathcal{N}(0, \sigma_w^2 I_d)$ with $\sigma_w > 0$, and ν_a, ν_b are bounded-support distributions on $\mathbb{R}$ (e.g., Rademacher) with $|a| \le A$ and $|b| \le B$ almost surely, and second moments $\sigma_a^2, \sigma_b^2 > 0$. The mean-field neural network is

$$P_q(x) := \int_{\mathbb{R}^{d+2}} a\, w\, \sigma(w^\top x + b)\, q(\mathrm{d}\theta), \tag{7}$$

which corresponds to the limit of $h(t_1)\, f_1(x;\theta) + x$ in (3). We minimize the DSM loss $L_t(q) := \frac{1}{2}\,\mathbb{E}_{x\sim p_t} \left\| f_1(x;q) - s^*(x,t) \right\|^2$ via the Wasserstein gradient flow restricted to the trainable direction:

$$\partial_s q_s \;=\; \nabla_w \cdot \left(q_s\, \nabla_w \frac{\delta L_t}{\delta q}(q_s) \right), \tag{8}$$

so the (a, b)-marginal of q_s remains $\nu_a \otimes \nu_b$ for all $s \ge 0$. The Stage 1 alignment risk is defined as

$$F_t(q) := \tfrac{1}{2}\,\mathbb{E}_{x\sim p_t} \left\| P_q(x) - \Pi_{\mathcal{M}}(x) \right\|^2. \tag{9}$$

Our analysis tracks F_t along this flow, rather than L_t directly, since F_t isolates the geometric alignment error.

Assumption 4.2 (Geometric Non-degeneracy)**.** There exists $\nu > 0$, *independent of* $h(t)$, such that for all $s \ge 0$,

$$\int_{\mathbb{R}^{d+2}} \left\| \nabla_w \frac{\delta F_t}{\delta q}(q_s)(\theta) \right\|^2 q_s(\mathrm{d}\theta) \;\ge\; \nu\, F_t(q_s). \tag{10}$$

Assumption 4.2 is a functional Polyak-Łojasiewicz inequality for F_t along the w-restricted Wasserstein flow, comparing the gradient norm of F_t at q_s to its functional value. Since (a, b) are frozen, this is effectively a PL condition on the w-marginal conditioned on (a, b). Assumption 4.2 ensures that any remaining projection error is detectable through the trainable feature directions w, conditional on the frozen labels (a, b). We verify it explicitly for linear manifolds in Proposition B.3.

Assumption 4.3 (Second-Moment Confinement)**.** The second moment of the trainable weights $m_2(q_s) := \int \|w\|^2\, q_s(\mathrm{d}\theta)$ is uniformly bounded along the flow: $\sup_{s\ge 0} m_2(q_s) \le M_2 < \infty$.

Assumption 4.3 is a standard regularity condition in mean-field analyses of two-layer networks [38, 11, 54], and we verify that it holds along the gradient flow under mild ℓ_2 regularization of w. Since (a, b) are frozen at bounded-support distributions, the corresponding moments $\int a^2 q_s = \sigma_a^2$ and $\int b^2 q_s = \sigma_b^2$ are constant in s, and all mixed moments involving a, b inherit the bounds $|a| \le A$, $|b| \le B$ automatically.

Remark 4.4 (On freezing a and b). Freezing the output coefficients a and biases b during the mean-field flow is a standard random-feature-style simplification in the analysis of two-layer networks [38, 11]; only the input weights w are subject to feature learning. Without this restriction, the mean-field network would be bilinear in (a, w), producing a coupled fourth-order moment $\mathbb{E}[a^2\|w\|^2]$ in the residual analysis that is not controlled by second moments of the marginals alone. Freezing (a, b) at bounded-support initializations sidesteps this issue; the expressivity required for collapse is then absorbed into the non-degeneracy condition (Assumption 4.2), which the convergence theorem (11) takes as a working hypothesis and which we verify explicitly for linear manifolds in Proposition B.3.

Theorem 4.5 (Dimensional Collapse)**.** *Assume $\mathcal{M} \subset \mathbb{R}^d$ is a compact C^∞ submanifold of intrinsic dimension k with reach $\tau > 0$, $p_0 \in C^2(\mathcal{M})$ with $0 < p_{\min} \le p_0 \le p_{\max} < \infty$ uniformly, and $\sigma \in C^2(\mathbb{R})$ is non-polynomial with $\max(\|\sigma\|_\infty, \|\sigma'\|_\infty, \|\sigma''\|_\infty) \le C_\sigma$. Fix $h(t_1) \le c\,\tau^2$ for a sufficiently small universal constant $c \in (0, 1)$. Under Assumptions 4.2 and 4.3, the Stage-1 alignment risk satisfies*

$$F_t(q_s) \;\le\; F_t(q_0)\,\exp\!\left(-\frac{\nu\, s}{2\,h(t_1)^2}\right) \;+\; \frac{C_t\,h(t_1)^2}{\nu}, \tag{11}$$

where $F_t(q_0) \le \frac{1}{2}\,\mathrm{Tr}(\Sigma_{\mathcal{M}}) + C_\sigma^2\,\sigma_a^2\,\sigma_w^2\,d$ with $\Sigma_{\mathcal{M}} := \mathbb{E}[\Pi_{\mathcal{M}}(x)\Pi_{\mathcal{M}}(x)^\top]$, and $C_t > 0$ depends on $\mathcal{M}$, τ, C_σ, A, and M_2 but not on s. Consequently, $F_t(q_s) \to C_t h(t_1)^2/\nu$ as $s \to \infty$, and the expected projection error satisfies $\mathbb{E}\|P_{q_s}(x) - \Pi_{\mathcal{M}}(x)\| = O(h(t_1))$.

Remark 4.6 (The role of the singularity). The exponential rate $\dfrac{\nu}{2\,h(t_1)^2}$ in (11) grows unboundedly as $h(t_1)$ shrinks within the admissible range $(0, c\tau^2]$: the $O(h(t_1)^{-1})$ singularity of the score function drives geometric learning rather than obstructing it, inducing exponentially fast collapse of the induced denoising map onto the data manifold projection. This complements the rate separation of [34]: both phenomena reflect the same score asymmetry at small noise, manifested statically as their $\Theta(h(t_1)^{-1})$ separation in the converged estimator and dynamically as the exponential rate above.

4.2 Stage 2: Density Estimation on the Manifold

Having established that Stage 1 produces an ϵ_{proj}-accurate projection $\hat{x} \approx \Pi_{\mathcal{M}}(x)$, we now analyze Stage 2. By construction of s_θ in (4), the singular normal component cancels up to the Stage-1 alignment error, and the optimization reduces to estimating the residual score $s^*_{\mathrm{res}}(x, t)$.

Definition 4.7. The on manifold principle residual is the restriction of its gradient on the manifold

$$\hat{s}^*(z) :=$$

We model f_2 as a Random Feature (RF) network with spatio-temporal features:

$$f_2(\hat{x}, t_2; U) = U\Phi(\hat{x}, t_2), \quad \Phi(\hat{x}, t_2) = \frac{1}{\sqrt{m}}\sigma\left(V_x^\top \hat{x} + b\right) \in \mathbb{R}^m, \tag{12}$$

where $V_x \in \mathbb{R}^{d\times m}$ and b are frozen random features and bias feature, and $U \in \mathbb{R}^{d\times m}$ is the only trainable parameter. Training Stage 2 thus reduces to a convex vector-valued ridge regression:

$$\hat{U} = \arg\min_U \frac{1}{n}\sum_{i=1}^{n}\left\|U\Phi(\hat{x}_{t_2}, t_2) - s^*_{\mathrm{res}}(\hat{x}_{t_2}, t_2)\right\|^2 + \lambda\|U\|_F^2. \tag{13}$$

Theorem 4.8 (Generalization of Stage 2)**.** *Fix two noise levels $h(t_1) \ll h(t_2)$, both in the manifold regime. Suppose further that the residual score satisfies the source condition $s^*_{\mathrm{res}} = T_K^\alpha g$ with $\|g\|_{L^2(\mathcal{M})} \le G$ and smoothness $\alpha > k/(2r)$, and let $\hat{U}$ be the regularized empirical-risk minimizer of* (13) *over n i.i.d. samples at noise level $h(t_2)$. Then with probability at least $1 - \delta$,*

$$\mathbb{E}_{x_{t_2}}\left\|s_{\hat{\theta}}(x_{t_2}, t_2) - s^*(x_{t_2}, t_2)\right\|^2 \le \underbrace{\frac{C_{\mathrm{int}}^2\log(1/\delta)}{n}}_{\text{estimation}} + \underbrace{O\big(m^{-(2\alpha r/k-1)}\big)}_{\text{approximation}} + \underbrace{C_{\mathrm{Lip}}^2\left(\frac{h(t_1)}{h(t_2)}\right)^2}_{\text{Stage 1 residual}}, \tag{14}$$

where the constants $C_{\mathrm{int}}, C_{\mathrm{Lip}} > 0$ depend on $(G, C_\sigma, \lambda, \mathcal{M})$ but not *on the ambient dimension d.*

Remark 4.9 (Two-timescale mechanism). The bound (14) reveals why the two-timescale structure is essential. Training Stage 1 at small $h(t_1)$ exploits the score singularity to drive $\epsilon_{\text{proj}} = O(h(t_1))$ (Theorem 4.5), while evaluating Stage 2 at moderate $h(t_2) \gg h(t_1)$ keeps the normal-direction amplification $1/h(t_2)$ bounded. The net effect on Stage 2 is the ratio $(h(t_1)/h(t_2))^2$, which can be driven to zero by widening the timescale gap independently of n or m.

Remark 4.10 (Comparison with existing rates). The intrinsic-dimensional dependence in (14) is in line with [10, 41, 55], but with two distinguishing features. First, while [10, 41] restrict to *linear* subspaces, Theorem 4.8 applies to general smooth manifolds via the projection $\hat{x}$ from Stage 1. Second, unlike [55, 8], which assume an oracle score, our bound is a generalization guarantee for the trained network on finite samples — and unlike the asymptotic random-feature curves of [21] that degrade on nonlinear manifolds, our two-stage decoupling preserves the intrinsic-dimensional rate.

4.3 End-to-End Sampling Guarantee

Theorems 4.5 and 4.8 bound the score estimation error within the manifold regime. We now lift these results to an end-to-end guarantee on the generated distribution by analyzing the reverse SDE over the full integration path. Following [15, 3], we partition $[t_{\min}, T]$ at $t_{\max}$ with $h(t_{\max}) \asymp \tau^2$. *Phase I* $[t_{\min}, t_{\max}]$ (*manifold regime*): p_t concentrates on the tubular neighborhood U_τ, the score exhibits the $O(h(t)^{-1})$ singularity, and the two-stage SiLD architecture (Eq. 4) applies. *Phase II* $[t_{\max}, T]$ (*Gaussian regime*): p_t no longer concentrates near $\mathcal{M}$, and the manifold-projection ansatz of Eq. 4 is no longer geometrically appropriate.

High-noise extension. To cover Phase II, we extend the score network with a complementary head, gated by a hard time indicator $\alpha(t) := 1[h(t) \le \tau^2]$:

$$s_\theta^{\text{full}}(x,t) = \alpha(t) \cdot s_\theta^{\text{SiLD}}(x,t) + (1-\alpha(t)) \cdot s_\theta^{\text{HN}}(x,t), \tag{15}$$

where s_θ^{SiLD} is the Stage 1+2 output of Eq. 4 and s_θ^{HN} is a multiplicatively time-modulated random-feature head:

$$s_\theta^{\text{HN}}(x,t) = \sum_{\ell=0}^{L-1} \phi_\ell(t) \cdot U_\ell \cdot \tfrac{1}{\sqrt{m}}\, \sigma\big(V_\ell^\top x\big)\,, \tag{16}$$

with $\{\phi_\ell\}_{\ell=0}^{L-1}$ a fixed Fourier basis on $[0,T]$, $V_\ell \in \mathbb{R}^{d\times m}$ frozen random features, and $U_\ell \in \mathbb{R}^{d\times m}$ the only trainable parameters, fitted by ridge regression on the DSM loss restricted to Phase II. The ansatz (16) is multiplicatively separable, so the induced kernel factorizes as $K_t \otimes K_x$ and the spatial eigenvalue analysis of Lemma B.6 extends without modification. At high noise, $s^*(x,t)$ is approximately affine in x with smooth time-dependent coefficients, so (16) achieves a parametric Phase II generalization bound that is *linear* in d rather than exponential, with a residual $O(e^{-T})$ accounting for the gap between p_T and $\mathcal{N}(0, I_d)$; see Lemma B.9 in Appendix B.5.

Theorem 4.11 (End-to-End Sampling Guarantee)**.** *Let p_{data} be supported on a k-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$, and let $p_{\text{gen},t_{\min}}$ be the distribution generated by simulating the reverse SDE with $s_{\hat{\theta}}^{\text{full}}$ from T down to $t_{\min}$. Choosing $t_{\min} \propto 1/n$ and $T \propto \log n$, with probability at least $1-\delta$:*

$$W_2\big(p_{\text{data}},\, p_{\text{gen},t_{\min}}\big) \;\lesssim\; \underbrace{\frac{\text{poly}(k)}{n^{1/4}}}_{\textit{Phase I (manifold)}} \;+\; \underbrace{\frac{\sqrt{C_{\text{HN}}\cdot d}}{n^{1/4}}}_{\textit{Phase II (Gaussian)}}\,, \tag{17}$$

where C_{HN} depends on $(\mathcal{M}, \sigma, L)$ but not on k.

Remark 4.12 (Comparison with existing rates). The end-to-end rate in (17) is best understood as a *training-level* guarantee, complementing prior *statistical* or *sampling-level* analyses. Sampling-complexity results [33, 28, 3] establish that DDPM samplers adapt to intrinsic dimension k given an oracle or sufficiently accurate score, but do not characterize how training produces such a score. Statistical-rate results on converged estimators [10, 41, 55, 2] obtain nonparametric Wasserstein rates of the form $n^{-c/(k+\text{const})}$ or $n^{-s/(2s+k)}$ on linear subspaces or smooth manifolds, but treat optimization abstractly. SiLD's contribution is orthogonal: we obtain a fixed-exponent $n^{-1/4}$ rate with an explicit account of how the induced denoising map collapse onto the data manifold projection $\Pi_{\mathcal{M}}$ under DSM gradient flow, and an architecture that confines the ambient dimension d to a *prefactor* $\sqrt{C_{\text{HN}}\, d}$ of the $n^{-1/4}$ rate rather than letting it appear in the exponent [41].

Algorithm 2 Practical Score-induced Latent Diffusion (Practical SiLD)

Require: Dataset $\mathcal{D}$; deterministic encoder E_ϕ and decoder G_ψ; latent diffusion network ϵ_θ; low-noise distribution q_{low}; latent diffusion schedule q_{lat}

Stage 1: Low-noise geometric representation learning

for each minibatch $x_0 \sim \mathcal{D}$ **do**
  Sample $t_1 \sim q_{\text{low}}$ and $\epsilon \sim \mathcal{N}(0, I)$
  $x_{t_1} \leftarrow \sqrt{\bar{\alpha}_{t_1}} x_0 + \sqrt{1 - \bar{\alpha}_{t_1}} \epsilon$
  $\hat{x}_0 \leftarrow G_\psi(E_\phi(x_{t_1}))$
  $\mathcal{L}_{\text{stage1}} \leftarrow \|\hat{x}_0 - x_0\|_2^2$
  Update (ϕ, ψ) using $\mathcal{L}_{\text{stage1}}$
end for
Freeze E_ϕ and G_ψ

Stage 2: Density learning in the learned coordinates

for each minibatch $x_0 \sim \mathcal{D}$ **do**
  $z_0 \leftarrow E_\phi(x_0)$
  Sample $t_2 \sim q_{\text{lat}}$ and $\xi \sim \mathcal{N}(0, I)$
  $z_{t_2} \leftarrow \sqrt{\bar{\alpha}_{t_2}} z_0 + \sqrt{1 - \bar{\alpha}_{t_2}} \xi$
  $\mathcal{L}_{\text{stage2}} \leftarrow \|\epsilon_\theta(z_{t_2}, t_2) - \xi\|_2^2$
  Update θ using $\mathcal{L}_{\text{stage2}}$
end for

Sampling

Sample $z_T \sim \mathcal{N}(0, I)$
Run the standard latent reverse diffusion process to obtain z_0
return $G_\psi(z_0)$

Remark 4.13 (Choice of $t_{\min}$). The choice $t_{\min} \propto 1/n$ optimally balances two opposing errors: the truncation gap $W_2(p_{\text{data}}, p_{\text{data},t_{\min}}) \lesssim \sqrt{h(t_{\min})}$ between the early-stopped and exact distributions, and the Girsanov integral on $[t_{\min}, t_{\max}]$, whose integrand grows as $h(t)^{-2}$ near $t_{\min}$.

5 Experiments

We evaluate SiLD (Algorithm 1) on synthetic low-dimensional manifolds and real-world image and molecular generation benchmarks. We compare SiLD against LDM [47], a standard latent diffusion model with a VAE encoder trained with KL regularization. All models share the same backbone and training budget to ensure a fair comparison. For images, we report Fréchet Inception Distance (FID) [26] for generation quality, and mean squared error (Recon MSE) and LPIPS [65] for reconstruction quality. For molecules, we report validity, uniqueness, novelty (fraction of valid generations absent from the training set), internal diversity (IntDiv), drug-likeness (QED) [4], and Fréchet ChemNet Distance (FCD) [45], alongside reconstruction MSE.

Toy experiment. We validate the dimensional-collapse mechanism on a synthetic mixture-of-Gaussians on a $k = 5$ linear subspace of $\mathbb{R}^{100}$, which admits an analytical score decomposition for direct comparison (setup and hyperparameters in Appendix C.1). Figure 1 shows the training dynamics: in Stage 1, the orthogonal contraction error (green) collapses rapidly while the manifold component error (red) remains flat; after the stage switch, Stage 2 exclusively refines the manifold component with W frozen. The sharp transition empirically confirms Theorem 4.5.

Stacked MNIST. We evaluate on Stacked MNIST, a 1000-mode benchmark that stacks three random MNIST digits into the R/G/B channels ($d = 2352$). The data manifold is non-smooth — discrete digit identities per channel — so the VAE's $\mathcal{N}(0, I)$ prior is misaligned with the data geometry, making this a diagnostic test for the Stage-1 objective. SiLD and LDM-CNN share an *identical* CNN encoder, decoder, and latent score network; only the Stage-1 loss differs. Both methods achieve full $1000/1000$ mode coverage (Table 1), ruling out mode collapse; at matched capacity, SiLD attains $2.0\times$ lower FID (7.94 ± 0.33 vs. 16.11 ± 0.09, mean$\pm$std over 3 seeds) and $1.77\times$ lower reconstruction MSE.

CelebA. Table 2 compares SiLD and LDM on CelebA (64×64) at matched architectural capacity, isolating the effect of the Stage-1 objective. At matched regularization strategy, SiLD consistently outperforms LDM: with MMD regularization, SiLD attains 42.52 FID vs. 43.58 for KL-regularized

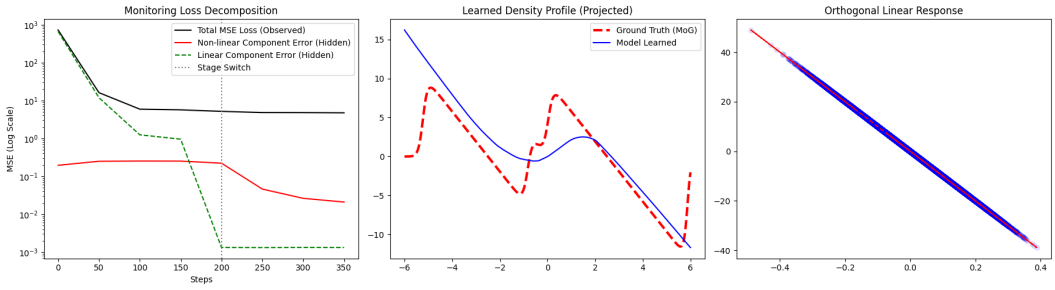


Figure 1: Two-stage learning dynamics on the Mixture of Gaussian on manifold experiment. *Left:* total MSE loss (black), manifold component error (red), and orthogonal contraction error (green, dashed); vertical dotted line marks the Stage 1 $\rightarrow$ Stage 2 switch. *Middle:* learned 1D score profile along the manifold matches the analytical MoG score after Stage 2. *Right:* learned score along normal directions matches the theoretical contraction $-x_\perp/h_t$, confirming correct isotropic behavior off the manifold.

Table 1: Stacked MNIST (1000-mode benchmark). SiLD and LDM-CNN share an identical CNN encoder, decoder, and latent score network; only the Stage-1 autoencoder objective differs. Both methods achieve full 1000/1000 mode coverage, so the $2.0\times$ FID gap and $1.77\times$ reconstruction-MSE gap reflect latent geometry rather than mode collapse.

Method	#modes / 1000	Recon MSE $\downarrow$	FID $\downarrow$
Real data	1000	–	–
LDM-CNN (β-VAE AE)	1000	$(6.28 \pm 0.08) \times 10^{-3}$	16.11 ± 0.09
SiLD (ours)	1000	$\mathbf{(3.55 \pm 0.02) \times 10^{-3}}$	$\mathbf{7.94 \pm 0.33}$

LDM; with GAN + EMA, SiLD reaches 39.57 vs. 40.69. Reconstruction MSE is lower for SiLD across every setting. The pure, unregularized SiLD variant attains the best reconstruction among KL-free methods (0.00238 vs. LDM's 0.00265) but higher FID (46.22), reflecting that the latent distribution can contain irregular structures absent any smoothing prior; light auxiliary regularization (MMD or GAN) recovers generation quality while preserving the reconstruction advantage. Similar trends hold on CelebA-HQ ($64 \times 64 \times 3$, intrinsic dimension $k_{95} = 223$); see Appendix C.4.

Molecular generation. We evaluate on four MoleculeNet benchmarks (QM9, MUV, HIV, PCBA) using the SELFIES representation [31], which guarantees syntactic validity by construction and thus isolates *distributional* learning as the object of comparison. SiLD and LDM share identical MLP encoder, decoder, and latent score network ($\sim$13M parameters); only the Stage-1 objective differs. We additionally compare against CharRNN, a 2-layer LSTM at matching scale, and report prior string-based methods on QM9 for external reference.

Table 3 reports the results. On QM9 (small molecules), SiLD and LDM both match the reference distribution closely, with SiLD slightly ahead on uniqueness (97.44% vs. 96.93%) and IntDiv (0.918 vs. 0.915). The comparison separates dramatically on the larger, drug-like datasets (HIV, MUV, PCBA): LDM undergoes distributional collapse, with uniqueness dropping to 0.96% on HIV, 95.60% on MUV, and 0.57% on PCBA, and IntDiv collapsing to 0.014–0.811 (versus real-data values of 0.86–0.90). In contrast, SiLD preserves the distribution: uniqueness 98.57%–99.42%, IntDiv 0.890–0.904, all within close range of the real training distribution. FCD shows the same pattern — on PCBA, SiLD achieves 17.79 versus LDM's 77.10 (real-data self-reference 9.12). CharRNN, the autoregressive baseline, matches the real-data distribution tightly but mostly by *memorizing*: novelty is 1.05% on QM9, 4.39% on HIV — it reproduces training molecules rather than generating new ones (except on the large PCBA where novelty reaches 64.8%). SiLD, in contrast, achieves near-perfect novelty (99.9%–100%) while preserving distributional structure, directly demonstrating that the Stage-1 objective produces a latent faithful to the data manifold rather than collapsing onto it or memorizing from it.

Table 2: Generation and reconstruction quality on CelebA (64×64). All models share identical encoder, decoder, and score network architecture. Only the autoencoder training objective differs.

Method	**AE Regularization**	**FID↓**	**Recon MSE↓**	**Eps Loss↓**
LDM (baseline)	KL divergence	43.58	0.00265	**0.283**
SiLD (ours)	none	46.22	0.00238	0.333
SiLD + MMD (ours)	MMD (α=0.1)	43.56	0.00229	0.353
SiLD + MMD (ours)	**MMD (α=1.0)**	**42.52**	**0.00210**	0.395
SiLD + MMD (ours)	MMD (α=5.0)	45.59	0.00207	0.424
LDM + GAN	GAN + EMA	40.69	0.00103	0.269
SiLD + GAN (ours)	GAN + EMA	**39.57**	**0.00098**	0.297

Table 3: Unconditional molecule generation results on four MoleculeNet benchmarks. All methods evaluated on the same protocol (10K samples, RDKit validity, Frechet ChemNet Distance against held-out test set). Our reproductions of prior methods use SELFIES representation for fair comparison. Best result per column per dataset in **bold** (among generative methods, excluding real data). ‡SELFIES guarantees syntactic validity. §CharRNN/Transformer AR achieve low FCD by partially memorizing training molecules (novelty <8% on QM9).

Dataset	**Model**	**Type**	**Repr.**	**Validity↑**	**Uniqueness↑**	**Novelty↑**	**Int. Div.↑**	**QED↑**	**FCD↓**
QM9	Real data	—	SELFIES	100.0%	96.18%	0.0%	0.919	0.466	5.05
	CharRNN§ [48]	RNN	SELFIES‡	**100.0%**	96.32%	1.05%	0.917	0.465	5.13
	Transformer AR§ [62, 40]	Transformer	SELFIES‡	**100.0%**	96.46%	5.26%	**0.920**	0.465	4.98
	VAE-only [22]	VAE prior	SELFIES‡	**100.0%**	84.43%	**88.93%**	0.900	0.439	7.43
	Latent GAN [46]	GAN in latent	SELFIES‡	**100.0%**	72.63%	74.58%	0.892	**0.474**	7.80
	LDM [47]	Latent diffusion	SELFIES‡	**100.0%**	96.93%	37.6%	0.915	0.472	**4.28**
	SiLD (ours)	Latent diffusion	SELFIES‡	**100.0%**	**97.44%**	35.9%	0.918	0.471	4.41
HIV	Real data	—	SELFIES	100.0%	88.93%	0.0%	0.898	0.513	3.24
	CharRNN§ [48]	RNN	SELFIES‡	99.97%	88.84%	4.39%	0.899	**0.514**	3.24
	Transformer AR§ [62, 40]	Transformer	SELFIES‡	99.93%	95.16%	60.29%	0.899	0.506	**2.91**
	VAE-only [22]	VAE prior	SELFIES‡	**100.0%**	94.18%	99.96%	**0.912**	0.388	17.66
	Latent GAN [46]	GAN in latent	SELFIES‡	**100.0%**	95.26%	99.99%	0.899	0.365	19.83
	LDM [47]	Latent diffusion	SELFIES‡	**100.0%**	0.96%	**100.0%**	0.014	0.114	60.91
	SiLD (ours)	Latent diffusion	SELFIES‡	**100.0%**	**98.74%**	99.9%	0.904	0.430	10.43
MUV	Real data	—	SELFIES	100.0%	95.03%	0.0%	0.860	0.675	4.52
	CharRNN§ [48]	RNN	SELFIES‡	**100.0%**	95.12%	8.17%	0.862	**0.676**	4.36
	Transformer AR§ [62, 40]	Transformer	SELFIES‡	**100.0%**	99.33%	83.08%	0.866	0.668	**4.06**
	VAE-only [22]	VAE prior	SELFIES‡	**100.0%**	94.98%	**100.0%**	**0.919**	0.364	33.48
	Latent GAN [46]	GAN in latent	SELFIES‡	**100.0%**	96.06%	**100.0%**	0.904	0.379	34.88
	LDM [47]	Latent diffusion	SELFIES‡	**100.0%**	95.60%	**100.0%**	0.811	0.237	41.20
	SiLD (ours)	Latent diffusion	SELFIES‡	**100.0%**	**99.42%**	**100.0%**	0.890	0.518	19.48
PCBA	Real data	—	SELFIES	100.0%	98.91%	0.0%	0.861	0.637	9.12
	CharRNN§ [48]	RNN	SELFIES‡	**100.0%**	99.53%	64.8%	0.864	**0.635**	8.08
	Transformer AR§ [62, 40]	Transformer	SELFIES‡	**100.0%**	**99.94%**	95.12%	0.869	0.624	**7.38**
	VAE-only [22]	VAE prior	SELFIES‡	**100.0%**	91.65%	99.74%	**0.915**	0.389	28.62
	Latent GAN [46]	GAN in latent	SELFIES‡	**100.0%**	94.93%	99.82%	0.898	0.357	30.41
	LDM [47]	Latent diffusion	SELFIES‡	**100.0%**	0.57%	94.7%	0.016	0.090	77.10
	SiLD (ours)	Latent diffusion	SELFIES‡	**100.0%**	98.57%	**99.8%**	0.895	0.491	17.79

6 Conclusion

We introduced *Score-induced Latent Diffusion (SiLD)*, a theoretically grounded framework in which manifold learning and density estimation emerge from a single DSM objective at different noise scales. The score singularity at small noise, rather than being an obstacle, drives a rapid dimensional collapse onto the data manifold; subsequent training at moderate noise reduces to ridge regression on a k-dimensional support, yielding an end-to-end sample complexity governed by the intrinsic dimension. Experiments on Stacked MNIST, CelebA, and four molecular benchmarks validate the predicted two-stage dynamic, most strikingly on drug-like molecular datasets, where KL-regularized LDMs undergo catastrophic distributional collapse while SiLD preserves the intrinsic distribution. More broadly, SiLD reframes the role of regularization in latent diffusion: the geometric structure of the score itself, rather than an externally imposed prior, is sufficient to organize the latent space.

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A Limitations

Our work is deliberately scoped to characterize the training dynamics of score matching when data lies on a low-dimensional manifold — the regime where the score singularity and intrinsic geometry interact most directly. The complementary problem of statistical optimality at the level of converged estimators, which has been well studied in prior work, is intentionally not the focus of our analysis. On the empirical side, we evaluate SiLD as a framework rather than as a system optimized for any single benchmark; integration with large-scale architectures and domain-specific design (e.g., text-to-image, video, scientific simulation) constitute natural extensions left to future work.

B Proofs of Main Results

We collect the proofs of all theoretical results in this appendix, organized in the order they appear in the main text.

B.1 Proofs of Proposition 3.1

Proof of Proposition 3.1. We reduce the VP marginal to a heat-kernel convolution on $\mathcal{M}$ via a rescaling, apply Laplace's method in Fermi coordinates, and return to x.

Step 1: Marginal density. Under the VP forward process, $x_t \mid x_0 = z \sim \mathcal{N}(a_t z, h(t) I_d)$ with $a_t := \sqrt{1 - h(t)}$, so

$$p_t(x) = (2\pi h(t))^{-d/2} \int_{\mathcal{M}} p_0(z)\, \exp\!\left(-\frac{\|x - a_t z\|^2}{2h(t)}\right) d\mathrm{Vol}_{\mathcal{M}}(z). \tag{18}$$

Step 2: Rescaling to a heat kernel on $\mathcal{M}$. Set $y := x/a_t$ and $\sigma^2 := h(t)/a_t^2$. Using $\|x - a_t z\|^2 = a_t^2 \|y - z\|^2$, equation (18) becomes

$$p_t(x) = a_t^{-d}\, q_{\sigma^2}(y), \qquad q_{\sigma^2}(y) := \frac{1}{(2\pi\sigma^2)^{d/2}} \int_{\mathcal{M}} p_0(z)\, e^{-\|y-z\|^2/(2\sigma^2)}\, d\mathrm{Vol}_{\mathcal{M}}(z).$$

Thus q_{σ^2} is the (Euclidean) heat-kernel convolution of p_0 on $\mathcal{M}$ at scale σ^2, and

$$\log p_t(x) = -d \log a_t + \log q_{\sigma^2}(x/a_t). \tag{19}$$

Since $\sigma^2 = h(t) + O(h(t)^2)$, the regime $h(t) \ll 1$ is equivalent to $\sigma^2 \ll 1$.

Step 3: Fermi coordinates on $\mathcal{M}$. Fix $y \in U_\tau$ and let $z_y^* := \Pi_{\mathcal{M}}(y)$, $\nu_y := y - z_y^* \in N_{z_y^*}\mathcal{M}$. Pick Fermi (geodesic normal) coordinates $u = (u^1, \dots, u^k)$ on $\mathcal{M}$ centered at z_y^*, with $\{e_1, \dots, e_k\}$ an orthonormal frame of $T_{z_y^*}\mathcal{M}$. The standard expansions are

$$z(u) = z_y^* + \sum_i e_i u^i + \tfrac{1}{2} \sum_{i,j} \mathrm{II}(e_i, e_j)\, u^i u^j + O(|u|^3), \tag{20}$$

$$\sqrt{\det g(u)} = 1 - \tfrac{1}{6}\mathrm{Ric}_{ij}(z_y^*)\, u^i u^j + O(|u|^3), \tag{21}$$

where $\mathrm{II} : T_{z_y^*}\mathcal{M} \times T_{z_y^*}\mathcal{M} \to N_{z_y^*}\mathcal{M}$ is the vector-valued second fundamental form. Because $\nu_y \perp T_{z_y^*}\mathcal{M}$ and II is normal-valued, a direct computation gives

$$\|y - z(u)\|^2 = \|\nu_y\|^2 + u^\top (I_k - B_{\nu_y})\, u + O(|u|^3), \qquad (B_{\nu_y})_{ij} := \langle \nu_y,\, \mathrm{II}(e_i, e_j) \rangle. \tag{22}$$

Positive definiteness of $I_k - B_{\nu_y}$ on U_τ follows from $\|B_{\nu_y}\|_{\mathrm{op}} \le \|\nu_y\|/\tau < 1$, since the principal curvatures of $\mathcal{M}$ are bounded by $1/\tau$.

Step 4: Laplace approximation of q_{σ^2}. Substituting (22) into the integral, expanding $p_0(z(u)) = p_0(z_y^*) + \nabla_{\mathcal{M}} p_0(z_y^*) \cdot u + O(|u|^2)$, and absorbing the volume-element correction into the $O(|u|^2)$ remainder, the first-order term in u vanishes by symmetry. The Gaussian integral $\int_{\mathbb{R}^k} e^{-u^\top (I_k - B_{\nu_y}) u/(2\sigma^2)}\, du = (2\pi\sigma^2)^{k/2} \det(I_k - B_{\nu_y})^{-1/2}$ then yields

$$q_{\sigma^2}(y) = (2\pi\sigma^2)^{-(d-k)/2}\, p_0(z_y^*)\, \frac{e^{-d_{\mathcal{M}}(y)^2/(2\sigma^2)}}{\det(I_k - B_{\nu_y})^{1/2}} \left(1 + O(\sigma^2)\right). \tag{23}$$

Step 5: Pullback to the x-variable. Combining (19) and (23), and collecting x-independent terms into $C(t)$,

$$\log p_t(x) = -\frac{d_{\mathcal{M}}(x/a_t)^2}{2\sigma^2} + \log p_0\big(\Pi_{\mathcal{M}}(x/a_t)\big) - \tfrac{1}{2}\log\det\big(I_k - B_{\nu_{x/a_t}}\big) + C(t) + O(h(t)). \quad (24)$$

We expand each term to leading order in $h(t)$. Using $1/a_t - 1 = h(t)/2 + O(h(t)^2)$ and Federer's identity $\nabla\big[\frac{1}{2}d^2_{\mathcal{M}}\big](x) = x - \Pi_{\mathcal{M}}(x) = \nu_x$,

$$d_{\mathcal{M}}(x/a_t)^2 = d_{\mathcal{M}}(x)^2 + h(t)\,\langle \nu_x,\, x\rangle + O(h(t)^2).$$

Combined with $1/\sigma^2 = 1/h(t) - 1 + O(h(t))$,

$$-\frac{d_{\mathcal{M}}(x/a_t)^2}{2\sigma^2} = -\frac{d_{\mathcal{M}}(x)^2}{2h(t)} + \tfrac{1}{2}d_{\mathcal{M}}(x)^2 - \tfrac{1}{2}\langle \nu_x, x\rangle + O(h(t)).$$

Using $x = \Pi_{\mathcal{M}}(x) + \nu_x$ and $\langle \nu_x, \nu_x\rangle = d_{\mathcal{M}}(x)^2$, the two $O(1)$ pieces collapse:

$$\tfrac{1}{2}d_{\mathcal{M}}(x)^2 - \tfrac{1}{2}\langle \nu_x, x\rangle = -\tfrac{1}{2}\langle \nu_x,\, \Pi_{\mathcal{M}}(x)\rangle.$$

The remaining two terms in (24) differ from their $a_t = 1$ counterparts by $O(h(t))$ (since $\Pi_{\mathcal{M}}$ and $\nu_\cdot$ are C^1 on U_τ and $x/a_t - x = O(h(t))$), so they contribute only to the remainder. Collecting:

$$\log p_t(x) = -\frac{d_{\mathcal{M}}(x)^2}{2h(t)} + \log p_0(\Pi_{\mathcal{M}}(x)) + H(\Pi_{\mathcal{M}}(x), \nu_x) + C(t) + O(h(t)), \quad (25)$$

where the $O(1)$ correction term is

$$H(z,\nu) := -\tfrac{1}{2}\langle \nu,\, z\rangle - \tfrac{1}{2}\log\det\big(I_k - B_\nu\big), \qquad z \in \mathcal{M},\ \nu \in N_z\mathcal{M}. \quad (26)$$

Crucially, H depends only on $(\Pi_{\mathcal{M}}(x),\, \nu_x)$, as required. The first summand in (26) is the VP-shrinkage contribution, while the second is the extrinsic-curvature correction.

Step 6: Taking the gradient. Applying ∇_x to (25) and using Federer's identity once more,

$$s^*(x,t) = -\frac{x - \Pi_{\mathcal{M}}(x)}{h(t)} + \nabla_x \log p_0\big(\Pi_{\mathcal{M}}(x)\big) + \nabla_x H\big(\Pi_{\mathcal{M}}(x),\, \nu_x\big) + O(h(t)),$$

which is the decomposition in the statement, with remainder $o(1)$ as $h(t) \to 0$. Term (I), $-(x - \Pi_{\mathcal{M}}(x))/h(t)$, is the conservative normal restoring force of order $O(h(t)^{-1})$; term (II), $\nabla \log p_0 + \nabla H$, is the $O(1)$ tangential score plus boundary corrections. □

B.2 Proof of Lemma 4.1

Proof of Lemma 4.1. Gradient identity. Since $\Psi'(z) = \sigma(z)$, the chain rule gives $\nabla_x \Psi(w_j^\top x + b_j) = \sigma(w_j^\top x + b_j)\, w_j$. Differentiating Φ_{net} termwise,

$$\nabla_x \Phi_{\mathrm{net}}(x;\theta) = \frac{x}{h(t)} - \frac{1}{h(t)}\sum_{j=1}^{m} a_j\, \sigma(w_j^\top x + b_j)\, w_j = \frac{x}{h(t)} - \frac{1}{h(t)}\, W\,\mathrm{diag}(a)\,\sigma(W^\top x + b) = -f_1(x;\theta),$$

which establishes the identity on all of $\mathbb{R}^d$.

Projection approximation. Define the target potential $\Phi^*(x,t) := d_{\mathcal{M}}(x)^2/(2\,h(t))$ and the smooth part

$$G^*(x) := \tfrac{1}{2}\|x\|^2 - \tfrac{1}{2}d_{\mathcal{M}}(x)^2 = \langle x, \Pi_{\mathcal{M}}(x)\rangle - \tfrac{1}{2}\|\Pi_{\mathcal{M}}(x)\|^2, \qquad x \in U_\tau,$$

where the second equality follows by expanding $d_{\mathcal{M}}(x)^2 = \|x - \Pi_{\mathcal{M}}(x)\|^2 = \|x\|^2 - 2\langle x, \Pi_{\mathcal{M}}(x)\rangle + \|\Pi_{\mathcal{M}}(x)\|^2$. Correspondingly, define the ridge sum

$$G_\theta(x) := \sum_{j=1}^{m} a_j\, \Psi(w_j^\top x + b_j),$$

so that $\Phi_{\text{net}}(x;\theta) = \|x\|^2/(2\,h(t)) - G_\theta(x)/h(t)$ and $\Phi^*(x,t) = \|x\|^2/(2\,h(t)) - G^*(x)/h(t)$. Taking gradients,

$$\nabla_x G_\theta(x) = \sum_{j=1}^{m} a_j\,\sigma(w_j^\top x + b_j)\,w_j = W\,\mathrm{diag}(a)\,\sigma(W^\top x + b), \qquad \nabla_x G^*(x) = \Pi_{\mathcal{M}}(x), \tag{27}$$

where the second identity follows from Federer's formula $\nabla_x[\frac{1}{2}d_{\mathcal{M}}(x)^2] = x - \Pi_{\mathcal{M}}(x)$ on U_τ [17]. Therefore

$$W\,\mathrm{diag}(a)\,\sigma(W^\top x + b) - \Pi_{\mathcal{M}}(x) = \nabla_x G_\theta(x) - \nabla_x G^*(x),$$

and (6) reduces to approximating G^* by ridge sums G_θ in the C^1-norm on $K := \overline{U_{\tau/2}}$.

The set K is compact since $\mathcal{M}$ is compact and K is the closed tube of radius $\tau/2$ around it. Under the standing assumption $\mathcal{M} \in C^\infty$, Federer's structure theorem gives $\Pi_{\mathcal{M}} \in C^\infty(U_\tau)$, hence $G^* \in C^\infty(K)$. In particular $G^* \in C^1(K)$.

Since $\sigma \in C^2$ is non-polynomial, so is its primitive $\Psi \in C^3$; indeed, if Ψ were polynomial then $\sigma = \Psi'$ would be polynomial, contradicting the hypothesis. The conservative-form parametrization $\theta = (W, a, b)$ produces precisely the standard ridge dictionary

$$\mathcal{R}(\Psi) := \Big\{ \textstyle\sum_{j=1}^{m} a_j\,\Psi(w_j^\top x + b_j) \;:\; m \in \mathbb{N},\; a_j \in \mathbb{R},\; w_j \in \mathbb{R}^d,\; b_j \in \mathbb{R} \Big\}$$

for the non-polynomial activation $\Psi \in C^3$. By the C^k density theorem for ridge functions [43, Thm. 4.1], $\mathcal{R}(\Psi)$ is dense in $C^1(K)$ under the norm $\|g\|_{C^1(K)} := \sup_K |g| + \sup_K \|\nabla g\|$. Consequently, for any $\delta > 0$ there exist $m \in \mathbb{N}$ and parameters $\theta = (W, a, b) \in \mathbb{R}^{d\times m} \times \mathbb{R}^m \times \mathbb{R}^m$ with $\|G_\theta - G^*\|_{C^1(K)} < \delta$. Choosing $\delta := \varepsilon$,

$$\sup_{x\in K} \|W\,\mathrm{diag}(a)\,\sigma(W^\top x + b) - \Pi_{\mathcal{M}}(x)\| = \sup_{x\in K} \|\nabla G_\theta(x) - \nabla G^*(x)\| \le \|G_\theta - G^*\|_{C^1(K)} < \varepsilon,$$

which is (6). □

B.3 Proof of Theorem 4.5

Proof of Theorem 4.5. **Step 1: Initial risk bound.** At $s = 0$, parameters are drawn from $q_0 = \nu_w \otimes \nu_a \otimes \nu_b$ with $\nu_w = \mathcal{N}(0, \sigma_w^2 I_d)$ and $|a| \le A$ almost surely under ν_a. By Jensen's inequality and $\|\sigma\|_\infty \le C_\sigma$,

$$\|P_{q_0}(x)\|^2 \;=\; \Big\| \int a\,w\,\sigma(w^\top x + b)\,q_0(\mathrm{d}\theta) \Big\|^2 \;\le\; \int a^2 \|w\|^2 \sigma(w^\top x + b)^2\,q_0(\mathrm{d}\theta) \;\le\; C_\sigma^2\,\sigma_a^2\,\sigma_w^2\,d,$$

where the last step uses independence of (w, a) under q_0: $\int a^2\|w\|^2\,q_0(\mathrm{d}\theta) = \sigma_a^2 \cdot \sigma_w^2 d$. This bound is independent of x, so by the triangle inequality and $\mathbb{E}\|\Pi_{\mathcal{M}}(x)\|^2 = \mathrm{Tr}(\Sigma_{\mathcal{M}})$,

$$F_t(q_0) = \tfrac{1}{2}\,\mathbb{E}\|P_{q_0}(x) - \Pi_{\mathcal{M}}(x)\|^2 \le \mathbb{E}\|P_{q_0}(x)\|^2 + \mathbb{E}\|\Pi_{\mathcal{M}}(x)\|^2 \le C_\sigma^2\,\sigma_a^2\,\sigma_w^2\,d + \mathrm{Tr}(\Sigma_{\mathcal{M}}).$$

Step 2: Loss decomposition. Let $R_q(x) := f_1(x;q) - s^*(x,t)$ be the score residual. By Proposition 3.1, $s^*(x,t) = -\frac{x - \Pi_{\mathcal{M}}(x)}{h(t)} + r(x,t)$ where $\|r(x,t)\| \le C_R$ uniformly on $U_{\tau/2}$ for a constant $C_R > 0$ depending on $\mathcal{M}$ and p_0. Since $f_1(x;q) = \frac{1}{h(t)}(P_q(x) - x)$, the residual is

$$R_q(x) = \frac{P_q(x) - \Pi_{\mathcal{M}}(x)}{h(t)} - r(x,t).$$

Squaring and taking expectations (using that p_t concentrates on $U_{\tau/2}$ up to an $e^{-\tau^2/(8h(t))}$ tail):

$$L_t(q) \;=\; \frac{1}{h(t)^2}\,F_t(q) \;-\; \frac{1}{h(t)}\,\widetilde{R}_t(q) \;+\; \tfrac{1}{2}\,\mathbb{E}\|r\|^2, \tag{28}$$

where $\widetilde{R}_t(q) := \mathbb{E}\big[\langle P_q(x) - \Pi_{\mathcal{M}}(x), r(x,t)\rangle\big]$ is the scale-crossing term. The first variation of L_t along the trainable w-direction satisfies

$$\nabla_w \frac{\delta L_t}{\delta q}(q)(\theta) \;=\; \frac{1}{h(t)^2}\,\nabla_w \frac{\delta F_t}{\delta q}(q)(\theta) \;-\; \frac{1}{h(t)}\,\nabla_w \frac{\delta \widetilde{R}_t}{\delta q}(q)(\theta). \tag{29}$$

Step 3: Energy dissipation. The chain rule along the w-restricted Wasserstein gradient flow (8) of L_t gives

$$\begin{aligned}\frac{d}{ds}F_t(q_s) &= -\int_{\mathbb{R}^{d+2}}\Big\langle \nabla_w \tfrac{\delta F_t}{\delta q},\ \nabla_w \tfrac{\delta L_t}{\delta q}\Big\rangle q_s(\mathrm{d}\theta)\\ &= -\frac{1}{h(t)^2}\int\Big\|\nabla_w \tfrac{\delta F_t}{\delta q}\Big\|^2 q_s(\mathrm{d}\theta)\ +\ \frac{1}{h(t)}\int\Big\langle \nabla_w \tfrac{\delta F_t}{\delta q},\ \nabla_w \tfrac{\delta \widetilde{R}_t}{\delta q}\Big\rangle q_s(\mathrm{d}\theta),\end{aligned}$$

where we used integration by parts (in w) for the w-restricted continuity equation and substituted (29).

Step 4: Young's inequality. For the cross-term, apply $\frac{1}{h(t)}\langle A, B\rangle \le \frac{1}{2h(t)^2}\|A\|^2 + \frac{1}{2}\|B\|^2$ pointwise, yielding

$$\frac{d}{ds}F_t(q_s)\ \le\ -\frac{1}{2h(t)^2}\int\Big\|\nabla_w \tfrac{\delta F_t}{\delta q}\Big\|^2 q_s(\mathrm{d}\theta)\ +\ \frac{1}{2}\int\Big\|\nabla_w \tfrac{\delta \widetilde{R}_t}{\delta q}\Big\|^2 q_s(\mathrm{d}\theta). \tag{30}$$

Step 5: Bound the remainder. The first variation of $\widetilde{R}_t$ at $\theta = (w, a, b)$ is $\frac{\delta \widetilde{R}_t}{\delta q}(\theta) = a\,\mathbb{E}_x\big[\sigma(w^\top x + b)\,\langle w, r(x,t)\rangle\big]$. Differentiating in w and using $|\sigma|, |\sigma'| \le C_\sigma$,

$$\nabla_w \frac{\delta \widetilde{R}_t}{\delta q}(\theta) = a\,\mathbb{E}_x\big[\sigma'(w^\top x + b)\,x\,\langle r(x,t), w\rangle + \sigma(w^\top x + b)\,r(x,t)\big],$$

so $\big\|\nabla_w \frac{\delta \widetilde{R}_t}{\delta q}(\theta)\big\| \le |a|\,C_\sigma\,C_R\,(\|w\|\cdot\mathbb{E}[\|x\|] + 1)$. Since $|a| \le A$ almost surely along the flow (a is frozen at ν_a), and by Assumption 4.3 $\int \|w\|^2 q_s(\mathrm{d}\theta) \le M_2$ uniformly in s,

$$\int\Big\|\nabla_w \tfrac{\delta \widetilde{R}_t}{\delta q}\Big\|^2 q_s(\mathrm{d}\theta)\ \le\ 2\,A^2\,C_\sigma^2\,C_R^2\big(\mathbb{E}[\|x\|^2]\,M_2 + 1\big)\ =:\ C_t,$$

where the inequality uses $(p+q)^2 \le 2p^2 + 2q^2$ followed by Jensen $(\mathbb{E}\|x\|)^2 \le \mathbb{E}\|x\|^2$. The constant $C_t > 0$ depends on $\mathcal{M}$, τ, C_σ, A, and M_2, but not on s.

Step 6: Non-degeneracy and Grönwall. Applying the Geometric Non-degeneracy Condition (10) to the first integral in (30) yields the linear differential inequality

$$\frac{d}{ds}F_t(q_s)\ \le\ -\frac{\nu}{2\,h(t)^2}\,F_t(q_s)\ +\ \frac{C_t}{2}.$$

Grönwall's lemma from time 0 to s gives

$$F_t(q_s)\ \le\ F_t(q_0)\,e^{-\nu s/(2h(t)^2)} + \frac{C_t\,h(t)^2}{\nu}\left(1 - e^{-\nu s/(2h(t)^2)}\right)\ \le\ F_t(q_0)\,e^{-\nu s/(2h(t)^2)} + \frac{C_t\,h(t)^2}{\nu},$$

establishing (11). Since $F_t(q_s) = \frac{1}{2}\mathbb{E}\|P_{q_s}(x) - \Pi_{\mathcal{M}}(x)\|^2$, the L^2 error of the learned projection converges to $O(h(t))$ as $s \to \infty$. □

Lemma B.1 (Second-Moment Confinement)**.** *Under the hypotheses of Theorem 4.5, suppose additionally that the training objective includes an ℓ_2 weight regularizer with coefficient $\lambda_w > 0$, i.e., the flow* (8) *is replaced by*

$$\partial_s q_s = \nabla_w \cdot \left(q_s\,\nabla_w \frac{\delta \widetilde{L}_t}{\delta q}(q_s)\right), \qquad \widetilde{L}_t(q) := L_t(q) + \frac{\lambda_w}{2}\int \|w\|^2\,q(dw). \tag{31}$$

Then the second moment satisfies

$$m_2(q_s)\ \le\ \max\big(m_2(q_0),\ C_{\mathrm{conf}}/\lambda_w\big) \qquad \textit{for all } s \ge 0, \tag{32}$$

where C_{conf} depends on C_σ, $\mathbb{E}[\|x\|^2]$, $h(t)$, and $\|s^*(\cdot,t)\|_{L^2(p_t)}$, but not on s.

Proof. The regularized velocity field is $v(w) = -\nabla_w \frac{\delta L_t}{\delta q}(w) - \lambda_w w$. Computing the evolution of the second moment,

$$\frac{d}{ds}m_2(q_s) = -2\int\big\langle w,\ \nabla_w \tfrac{\delta L_t}{\delta q}(w)\big\rangle\,q_s(dw)\ -\ 2\lambda_w\,m_2(q_s). \tag{33}$$

For the first term, the functional derivative of L_t is $\frac{\delta L_t}{\delta q}(w) = \frac{1}{h(t)}\,\mathbb{E}_x\big[\sigma(w^\top x)\,\langle R_q(x),\,w\rangle\big]$ where $R_q(x) := f_1(x;q) - s^*(x,t)$ is the residual. Its w-gradient satisfies

$$\nabla_w \frac{\delta L_t}{\delta q}(w) = \frac{1}{h(t)}\,\mathbb{E}_x\big[\sigma'(w^\top x)\,x\,\langle R_q, w\rangle + \sigma(w^\top x)\,R_q\big].$$

Taking the inner product with w and using $|\sigma| \le C_\sigma$, $|\sigma'| \le C_\sigma$:

$$\big|\big\langle w,\,\nabla_w \tfrac{\delta L_t}{\delta q}(w)\big\rangle\big| \;\le\; \frac{C_\sigma}{h(t)}\,\big(\|w\|^2 + \|w\|\big)\,\mathbb{E}\big[\|R_q(x)\|\,(\|x\|+1)\big].$$

Since $\|R_q\|_{L^2}^2 = 2L_t(q_s)$ is non-increasing along the gradient flow of $\widetilde{L}_t$, and $\mathbb{E}[(\|x\|+1)^2]$ is a fixed constant $C_{x,t}^2$ under p_t, integrating against q_s yields

$$\left|\int\big\langle w,\,\nabla_w \tfrac{\delta L_t}{\delta q}\big\rangle\,q_s(dw)\right| \;\le\; \frac{C_\sigma C_{x,t}\sqrt{2L_t(q_0)}}{h(t)}\,\big(m_2(q_s) + \sqrt{m_2(q_s)}\big) \;\le\; C'(m_2(q_s)+1),$$

where $C' := C_\sigma C_{x,t}\sqrt{2L_t(q_0)}/h(t)$. Substituting back into (33):

$$\frac{d}{ds}m_2 \;\le\; 2C'(m_2+1) - 2\lambda_w m_2 = -(2\lambda_w - 2C')\,m_2 + 2C'.$$

Choosing $\lambda_w > C'$ (i.e., λ_w depends on $h(t)$, C_σ, and $L_t(q_0)$), the coefficient of m_2 is negative, and Grönwall's lemma gives (32) with $C_{\mathrm{conf}} = 2C'$. □

Remark B.2. Since Stage 1 operates at a fixed noise level $h(t_1)$, the regularization coefficient λ_w is a fixed hyperparameter chosen before training. The regularization introduces a bias of order $O(\lambda_w)$ in the alignment risk, which can be made arbitrarily small relative to the $O(h(t)^2)$ residual in (11) by choosing $\lambda_w = o(h(t)^2)$.

Proposition B.3 (Non-degeneracy for Linear Manifolds)**.** *Let $\mathcal{M} = \mathrm{range}(A)$ for some $A \in \mathbb{R}^{d\times k}$ with $A^\top A = I_k$, so that $\Pi_{\mathcal{M}}(x) = AA^\top x$. Let $\Sigma_t := \mathbb{E}_{x\sim p_t}[xx^\top] = (1-h(t))\Sigma_0 + h(t)I_d$, where $\Sigma_0 = \mathbb{E}[x_0 x_0^\top]$. Assume $\sigma \in C^2(\mathbb{R})$ is non-polynomial with $\|\sigma\|_\infty, \|\sigma'\|_\infty \le C_\sigma$, the trainable weights satisfy $\int \|w\|^2 q_s(\mathrm{d}\theta) \le M_2$ for all $s \ge 0$, and the frozen output coefficients have second moment $\sigma_a^2 := \mathbb{E}_{\nu_a}[a^2] > 0$.*

Then the Geometric Non-degeneracy Condition (10) *holds with*

$$\nu \;=\; \sigma_a^2 \,\cdot\, \lambda_{\min}(\mathcal{T}_t) \,\cdot\, \mu_\sigma^2, \tag{34}$$

where $\lambda_{\min}(\mathcal{T}_t)$ is the smallest eigenvalue of the feature covariance operator

$$\mathcal{T}_t[g](w,b) := \mathbb{E}_{x\sim p_t}\big[\sigma(w^\top x + b)\,\langle g, x\rangle\,x\big],$$

acting on vector fields aligned with $(\mathrm{range}(A))^\perp$ and indexed by (w,b) under the (w,b)-marginal of q_s, and $\mu_\sigma > 0$ is a constant depending only on σ and the spectral bounds of Σ_t. The factor σ_a^2 originates from the multiplicative a in $\nabla_w \frac{\delta F_t}{\delta q}(\theta)$ (cf. Step 5 of the proof of Theorem 4.5).

In particular, ν is independent of $h(t)$ in the following sense: for any compact interval $[h_{\min}, h_{\max}] \subset (0,1)$,

$$\inf_{h(t)\in[h_{\min},h_{\max}]} \nu(h(t)) \;>\; 0, \tag{35}$$

since Σ_t is positive definite and varies continuously in $h(t)$, and the spectral gap of $\mathcal{T}_t$ depends continuously on Σ_t.

Proof sketch. The argument follows the three-step structure of the original case (w-only flow), adapted to the Option B setting in which (a,b) are frozen at $\nu_a \otimes \nu_b$ and only w evolves.

Step 1: Reduction to a kernel eigenvalue problem. For the linear manifold, $\Delta_q(x) := P_q(x) - AA^\top x$ and $F_t(q) = \frac{1}{2}\mathbb{E}\|\Delta_q(x)\|^2$. The first variation along the trainable direction is

$$\nabla_w \frac{\delta F_t}{\delta q}(\theta) \;=\; a\,\mathbb{E}_x\big[\sigma'(w^\top x + b)\,x\,\langle \Delta_q, w\rangle + \sigma(w^\top x + b)\,\Delta_q\big],$$

so squaring produces an outer a^2 factor. Since a is frozen at ν_a (independent of (w, b) at $s = 0$ and unchanged thereafter) and $\sigma_a^2 = \mathbb{E}[a^2] > 0$, this factor contributes σ_a^2 to ν. The residual (w, b)-integral then reduces to the spectral gap of the integral operator with kernel

$$K_t\big((w,b),(w',b')\big) := \mathbb{E}_{x\sim p_t}\big[\sigma(w^\top x + b)\,\sigma(w'^\top x + b')\,(w^\top w')\big]$$

acting on L^2 of the (w, b)-marginal of q_s.

Step 2: Spectral gap from non-degeneracy of σ. Since σ is non-polynomial and C^2, the family $\{x \mapsto \sigma(w^\top x + b) : (w, b) \in \mathrm{supp}(\nu_w \otimes \nu_b)\}$ spans a dense subspace of $L^2(p_t)$ (by the non-polynomial universality theorem for ridge functions, with translation parameter b). Combined with the positive definiteness of Σ_t for all $h(t) > 0$, this ensures $\lambda_{\min}(\mathcal{T}_t) > 0$. Concretely, for the Gaussian distribution p_t with covariance Σ_t, the Hermite expansion of $\sigma(\cdot + b)$ has non-vanishing coefficients at all orders (since σ is non-polynomial), and each Hermite component contributes a strictly positive term to the spectral decomposition of K_t.

Step 3: Uniformity in $h(t)$. The eigenvalues of K_t are continuous functions of Σ_t (since they are expectations of continuous functions of the data covariance). On the compact set $h(t) \in [h_{\min}, h_{\max}]$, Σ_t ranges over a compact family of positive definite matrices, so $\lambda_{\min}(\mathcal{T}_t)$ attains a positive minimum. Therefore ν can be chosen uniformly over $h(t) \in [h_{\min}, h_{\max}]$. □

Remark B.4 (Scope of the non-degeneracy condition). Proposition B.3 verifies the non-degeneracy condition (10) for linear manifolds with ν explicitly independent of $h(t)$, confirming that the exponential rate $\nu/(2h(t)^2)$ in (11) grows unboundedly as $h(t_1)$ shrinks within the admissible range $(0, c\tau^2]$. For general nonlinear manifolds $\mathcal{M}$, we retain (10) as an assumption; extending the verification to curved manifolds via a perturbative argument (treating curvature corrections as lower-order perturbations of the linear spectral gap) is an interesting direction for future work.

B.4 Proof of Theorem 4.8

Bounded feature energy. A crucial property of the RF network is that its feature energy is entirely independent of the ambient dimension d.

Lemma B.5 (Bounded Spatio-Temporal Feature Energy)**.** *Assume the activation $\sigma \in C^2(\mathbb{R})$ satisfies $\|\sigma\|_\infty \le C_\sigma$ for some constant $C_\sigma > 0$. Then for any $V_x \in \mathbb{R}^{d\times m}$ and b, the random feature map $\Phi(\hat{x}, t) := \frac{1}{\sqrt{m}}\,\sigma\big(V_x^\top \hat{x} + b\big) \in \mathbb{R}^m$ satisfies*

$$\sup_{\hat{x}\in\mathcal{M},\,t} \|\Phi(\hat{x},t)\|^2 \le C_\sigma^2 =: C_{ST}, \tag{36}$$

where C_{ST} depends only on σ and is independent of the ambient dimension d, the feature dimension m, the manifold $\mathcal{M}$, and the diffusion time t.

Kernel eigenvalue decay. The generalization of Stage 2 hinges on the spectral properties of the induced kernel $K(\hat{x}, \hat{x}') = \mathbb{E}_V[\sigma(V^\top \hat{x})\sigma(V^\top \hat{x}')]$ restricted to $\mathcal{M}$. Since $\mathcal{M}$ is a compact k-dimensional Riemannian manifold, Weyl's law controls the eigenvalue decay.

Lemma B.6 (Eigenvalue Decay on Manifold)**.** *Let $(\mathcal{M}, g)$ be a compact k-dimensional C^∞ Riemannian manifold. Consider the kernel*

$$K(\hat{x},\hat{x}') := \mathbb{E}_V\big[\sigma(V^\top \hat{x})\,\sigma(V^\top \hat{x}')\big], \tag{37}$$

where V is drawn from a distribution with density on $\mathbb{R}^{d\times m}$, and assume $\sigma \in C^r(\mathbb{R})$ for some $r \ge 2$ with bounded derivatives up to order r. Then:

1. ***(Kernel regularity.)*** *The kernel K belongs to $C^r(\mathcal{M}\times\mathcal{M})$, hence to $H^{r-k/2}(\mathcal{M}\times\mathcal{M})$ by Sobolev embedding.*
2. ***(Eigenvalue decay.)*** *The eigenvalues $\{\lambda_j\}_{j\ge1}$ of the induced integral operator $T_K : L^2(\mathcal{M}) \to L^2(\mathcal{M})$ satisfy*

$$\lambda_j \;\le\; C_{\mathcal{M},r}\,\|K\|_{C^r}\; j^{-r/k}, \tag{38}$$

where $C_{\mathcal{M},r}$ depends only on $(\mathcal{M}, g)$ and r, not on the ambient dimension d.

The polynomial decay rate $j^{-r/k}$ depends on the *intrinsic* dimension k rather than d, which is the key to controlling the ambient dependence in Stage 2.

Proof. By definition of Φ,

$$\|\Phi(\hat{x},t)\|^2 = \frac{1}{m}\sum_{i=1}^{m}\sigma\big((V_x^\top \hat{x})_i + b_i\big)^2 \le \frac{1}{m}\sum_{i=1}^{m}\|\sigma\|_\infty^2 = \|\sigma\|_\infty^2 \le C_\sigma^2,$$

where the inequality applies the pointwise bound $|\sigma(z)| \le C_\sigma$ termwise. The resulting bound is independent of $(\hat{x},t)$, (V_x,V_t), and m, so taking the supremum over $\hat{x}\in\mathcal{M}$ and t in the time interval under consideration yields (36). □

Proof. **Part (i): Kernel regularity.** Since $\sigma \in C^r(\mathbb{R})$ with bounded derivatives, the map $\hat{x} \mapsto \sigma(V^\top\hat{x})$ is C^r on $\mathcal{M}$ for each fixed V, with derivatives bounded uniformly in V by $\max_{0\le j\le r}\|\sigma^{(j)}\|_\infty \cdot \|V\|_{\mathrm{op}}^j$. Since the distribution of V has finite moments of all orders, dominated convergence ensures that $K(\hat{x},\hat{x}') = \mathbb{E}_V[\sigma(V^\top\hat{x})\sigma(V^\top\hat{x}')]$ inherits C^r regularity jointly in $(\hat{x},\hat{x}')\in\mathcal{M}\times\mathcal{M}$. The Sobolev embedding $C^r(\mathcal{M}) \hookrightarrow H^\tau(\mathcal{M})$ for $\tau \le r - k/2$ then gives $K \in H^\tau(\mathcal{M}\times\mathcal{M})$.

Part (ii): Eigenvalue decay. The argument proceeds via the factorization of T_K through Sobolev spaces.

Step 1: Mapping property. Since $K(\cdot,\hat{x}')\in C^r(\mathcal{M})$ uniformly in $\hat{x}'$, the integral operator T_K maps $L^2(\mathcal{M})$ boundedly into $C^r(\mathcal{M}) \hookrightarrow H^r(\mathcal{M})$:

$$\|T_K f\|_{H^r(\mathcal{M})} \;\le\; C\,\|K\|_{C^r(\mathcal{M}\times\mathcal{M})}\,\|f\|_{L^2(\mathcal{M})}.$$

Step 2: Singular values of the Sobolev embedding. The inclusion $\iota : H^r(\mathcal{M}) \hookrightarrow L^2(\mathcal{M})$ is compact, and its singular values $\{\mu_j\}_{j\ge1}$ satisfy $\mu_j \asymp j^{-r/k}$. This follows from the spectral theory of the Laplace–Beltrami operator $-\Delta_{\mathcal{M}}$ on the compact manifold $\mathcal{M}$: Weyl's law gives that the eigenvalues of $-\Delta_{\mathcal{M}}$ grow as $\lambda_j^\Delta \asymp j^{2/k}$, so the eigenvalues of $(I-\Delta_{\mathcal{M}})^{-r/2}$ (which characterize $H^r \hookrightarrow L^2$) decay as $j^{-r/k}$.

Step 3: Factorization. Since $T_K = \iota\circ(T_K : L^2 \to H^r)$, the multiplicative property of singular values gives

$$\lambda_j(T_K) \;\le\; \|T_K : L^2 \to H^r\|_{\mathrm{op}} \;\cdot\; \mu_j \;\le\; C\,\|K\|_{C^r}\; j^{-r/k},$$

which is (38). The constant depends on $(\mathcal{M},g)$ through the Weyl constant and the Sobolev embedding constant, but not on the ambient dimension d. □

Remark B.7 (Effective decay rate). Under the standing assumption $\sigma\in C^2(\mathbb{R})$, Lemma B.6 gives $\lambda_j = O(j^{-2/k})$. If σ is C^∞ (e.g., $\sigma=\tanh$ or sigmoid), the decay $\lambda_j = O(j^{-r/k})$ holds for arbitrarily large r, yielding faster-than-any-polynomial decay. The polynomial rate $j^{-r/k}$ depends on the intrinsic dimension k rather than the ambient dimension d, which is the key to breaking the curse of dimensionality in Stage 2.

Proof of Theorem 4.8. The proof proceeds in four steps.

Step (i): Rademacher complexity (estimation error). Consider the hypothesis class $\mathcal{F}_R = \{x \mapsto U\Phi(x,t) : \|U\|_F \le B_U\}$. By Lemma B.5, $\|\Phi(\hat{x},t)\|^2 \le C_\sigma^2$ independently of d, m, $\mathcal{M}$, and t. The empirical Rademacher complexity of $\mathcal{F}_R$ is therefore bounded by

$$\hat{\mathcal{R}}_n(\mathcal{F}_R) \;\le\; \frac{B_U}{\sqrt{n}}\sqrt{\frac{1}{n}\sum_{i=1}^{n}\|\Phi(\hat{x}_i,t_i)\|^2} \;\le\; \frac{B_U\,C_\sigma}{\sqrt{n}} \;=\; \frac{C_{\mathrm{int}}}{\sqrt{n}}, \tag{39}$$

independently of d.

It remains to show that B_U is d-independent. The ridge regression solution is $\hat{U} = \mathbb{E}[s_{\mathrm{res}}^*\,\Phi^\top](\mathbb{E}[\Phi\,\Phi^\top]+\lambda I)^{-1}$. By the Cauchy–Schwarz inequality,

$$\|\hat{U}\|_F \;\le\; \frac{1}{\lambda}\,\big\|\mathbb{E}[s_{\mathrm{res}}^*\,\Phi^\top]\big\|_F \;\le\; \frac{1}{\lambda}\,\sqrt{\mathbb{E}\|s_{\mathrm{res}}^*\|^2}\;\cdot\;\sqrt{\mathbb{E}\|\Phi\|^2} \;\le\; \frac{\|s_{\mathrm{res}}^*\|_{L^2}\,C_\sigma}{\lambda}.$$

The key observation is that $s^*_{\mathrm{res}}(\hat{x}, t) \in T_{\hat{x}}\mathcal{M}$ is a tangential vector field on $\mathcal{M}$, so its L^2-norm is controlled by intrinsic quantities: under the source condition, $\|s^*_{\mathrm{res}}\|_{L^2} \le \|T_K^s\|_{\mathrm{op}} \cdot G \le \lambda_1^s\, G$, where λ_1 is the leading eigenvalue of T_K. Therefore $B_U \le \lambda_1^s\, G\, C_\sigma/\lambda$, which depends on $(\mathcal{M}, p_0, \sigma, \lambda)$ but not on d.

Standard Rademacher-to-generalization conversion with a union bound over the time discretization then gives the estimation error $C_{\mathrm{int}}^2 \log(1/\delta)/n$.

Step (ii): Approximation error. By Lemma B.6 and the source condition, the Random Feature approximation error is controlled by the spectral tail:

$$\epsilon_{\mathrm{approx}}(m, k) \;=\; \sum_{j>cm} \lambda_j^{2\alpha} \;\le\; C_\lambda^{2\alpha} \sum_{j>cm} j^{-2\alpha r/k} \;=\; O\big(m^{-(2\alpha r/k - 1)}\big),$$

where the convergence of the sum uses $2\alpha r/k > 1$. The decay rate depends only on k (through the eigenvalue exponent r/k) and not on d.

Step (iii): Stage 1 error propagation. The imperfect projection $\hat{x} \approx \Pi_{\mathcal{M}}(x)$ introduces two distinct error contributions. By the score architecture (1),

$$s_\theta(x,t) - s^*(x,t) = \underbrace{\frac{\Pi_{\mathcal{M}}(x) - \hat{x}}{h(t)}}_{\text{(a) normal mismatch}} + \underbrace{\big(f_2(\hat{x},t) - s^*_{\mathrm{res}}(\hat{x},t)\big)}_{\text{Stage 2 error at } \hat{x}} + \underbrace{\big(s^*_{\mathrm{res}}(\hat{x},t) - s^*_{\mathrm{res}}(\Pi_{\mathcal{M}}(x),t)\big)}_{\text{(b) tangential feature perturbation}}.$$

Term (a): Normal mismatch. This is the dominant Stage 1 contribution. Taking the squared expectation:

$$\mathbb{E}\left\|\frac{\Pi_{\mathcal{M}}(x) - \hat{x}}{h(t)}\right\|^2 = \frac{\epsilon_{\mathrm{proj}}^2}{h(t)^2}.$$

Term (b): Tangential feature perturbation. Since s^*_{res} is a C^1 vector field on $\mathcal{M}$ with Lipschitz constant $\mathrm{Lip}(s^*_{\mathrm{res}})$ depending on the curvature and density gradients of $\mathcal{M}$:

$$\mathbb{E}\big\|s^*_{\mathrm{res}}(\hat{x},t) - s^*_{\mathrm{res}}(\Pi_{\mathcal{M}}(x),t)\big\|^2 \;\le\; \mathrm{Lip}(s^*_{\mathrm{res}})^2\, \epsilon_{\mathrm{proj}}^2.$$

Cross term. The cross term between (a) and the Stage 2 error is controlled by the AM–GM inequality: $2\langle A, B\rangle \le \|A\|^2 + \|B\|^2$, so it can be absorbed into the other terms at the cost of constant factors.

Combining, the total Stage 1 error is

$$\underbrace{\frac{\epsilon_{\mathrm{proj}}^2}{h(t)^2}}_{\text{dominant}} + \underbrace{\mathrm{Lip}(s^*_{\mathrm{res}})^2\, \epsilon_{\mathrm{proj}}^2}_{\text{lower order since } h(t) \ll 1} \;\le\; \frac{C_{\mathrm{Lip}}^2\, \epsilon_{\mathrm{proj}}^2}{h(t)^2},$$

where $C_{\mathrm{Lip}} := \sqrt{1 + h(t)^2\, \mathrm{Lip}(s^*_{\mathrm{res}})^2} \ge 1$ absorbs both contributions.

Step (iv): Combining. Summing the three terms from Steps (i)–(iii) yields (14). □

B.5 Proof of Theorem 4.11

This section establishes the Phase II generalization bound and the end-to-end Wasserstein-2 sampling guarantee. The argument proceeds in three pieces: an affine structural decomposition of the high-noise score (Lemma B.8), a parametric Phase II generalization bound for the multiplicative random-feature head (Lemma B.9), and the Girsanov-based end-to-end argument that combines Phase I and Phase II.

High-noise score structure. A crucial property of the score function in the Gaussian regime is that it is approximately *affine* in x, with smooth time-dependent coefficients. This linear-in-x structure is what enables the multiplicative random-feature ansatz (16) to achieve a parametric (rather than nonparametric) generalization bound.

Lemma B.8 (Affine approximation of the high-noise score)**.** *Assume $\mathcal{M}$ has finite diameter $R_{\mathcal{M}}$. For any t with $h(t) \in [\tau^2, 1)$, write $a_t := \sqrt{1 - h(t)}$. Then the score function admits the decomposition*

$$s^*(x,t) = -\frac{x}{h(t)} + \frac{a_t}{h(t)}\mu_t + \frac{a_t^2}{h(t)^2} C_t\, x + R_t(x), \tag{40}$$

where $\mu_t := \mathbb{E}_{z\sim p_0}[z] \in \mathbb{R}^d$, $C_t := \mathrm{Cov}_{z\sim p_0}(z) \in \mathbb{R}^{d\times d}$ are the first two moments of p_0, and the residual satisfies

$$\sup_{x\in\mathrm{supp}(p_t)} \|R_t(x)\| \;\le\; C_{\mathcal{M}}^{(1)} \cdot \frac{a_t^3}{h(t)^3} \cdot R_{\mathcal{M}}^3, \tag{41}$$

for a universal constant $C_{\mathcal{M}}^{(1)} > 0$.

Proof. The marginal density satisfies

$$p_t(x) = (2\pi h(t))^{-d/2} \exp\!\left(-\frac{\|x\|^2}{2h(t)}\right) \cdot \Phi_t(x), \qquad \Phi_t(x) := \mathbb{E}_{z\sim p_0}\left[\exp\!\left(\frac{a_t\langle x, z\rangle}{h(t)} - \frac{a_t^2\|z\|^2}{2h(t)}\right)\right]. \tag{42}$$

Taking gradients,

$$s^*(x,t) = -\frac{x}{h(t)} + \nabla_x \log \Phi_t(x). \tag{43}$$

Setting $\xi := a_t x/h(t)$, Φ_t is the moment generating function of the tilted distribution $p_0(z)\exp(-a_t^2\|z\|^2/(2h(t)))$ evaluated at ξ. Cumulant expansion of $\log\Phi_t$ around $\xi = 0$ gives

$$\log \Phi_t(\xi) = \langle \mu_t, \xi\rangle + \tfrac{1}{2}\xi^\top C_t \xi + O(\|\xi\|^3), \tag{44}$$

with cubic remainder bounded by the third moment of p_0, which is in turn bounded by $R_{\mathcal{M}}^3$ since p_0 is supported on $\mathcal{M} \subset B(0, R_{\mathcal{M}})$. Differentiating in x and substituting $\xi = a_t x/h(t)$ yields (40) with the cubic remainder controlled as in (41). □

Phase II generalization. The kernel induced by (16) factorizes as $K_t \otimes K_x$ due to the multiplicative structure. The spatial component K_x inherits the eigenvalue decay of Lemma B.6 on $\mathcal{M}$ (or, in Phase I, on a bounded ball in $\mathbb{R}^d$), while the temporal component K_t is finite-dimensional (L Fourier modes). The combination yields a hypothesis class whose Rademacher complexity is parametric in n with a d-dependent norm scale that propagates linearly into the bound.

Lemma B.9 (Phase II generalization)**.** *Assume $\mathcal{M}$ has finite diameter $R_{\mathcal{M}}$, the activation $\sigma \in C^2$ satisfies $\|\sigma\|_\infty, \|\sigma'\|_\infty \le C_\sigma$, and the Fourier basis $\{\phi_\ell\}_{\ell=0}^{L-1}$ satisfies $L \ge L_0$ for a constant L_0 depending only on the noise schedule. Let $\hat{U} = (\hat{U}_0, \dots, \hat{U}_{L-1})$ be the regularized empirical-risk minimizer*

$$\hat{U} \;=\; \arg\min_{U} \frac{1}{n}\sum_{i=1}^{n} \left\| s_\theta^{\mathrm{HN}}(x_{t_i}, t_i) - s^*(x_{t_i}, t_i)\right\|^2 + \lambda \sum_{\ell} \|U_\ell\|_F^2, \tag{45}$$

over n i.i.d. samples (x_{t_i}, t_i) with $t_i \sim \mathrm{Unif}([t_{\max}, T])$ and $x_{t_i} \sim p_{t_i}$. Then with probability at least $1-\delta$,

$$\mathbb{E}_{t\in[t_{\max},T]}\, \mathbb{E}_{x_t\sim p_t} \left\| s_{\hat{\theta}}^{\mathrm{HN}}(x_t, t) - s^*(x_t, t)\right\|^2 \;\le\; \frac{C_{\mathrm{HN}} \cdot d \cdot \log(1/\delta)}{n} + O(e^{-T}), \tag{46}$$

where C_{HN} depends on $(R_{\mathcal{M}}, C_\sigma, L, \lambda)$ but not on the intrinsic dimension k.

Proof. The proof proceeds in three steps: approximation, estimation, and the residual at T.

Step 1 (Approximation). By Lemma B.8, the dominant terms of $s^*(x,t)$ in Phase I form the affine target

$$s_{\mathrm{aff}}^*(x,t) := -\frac{x}{h(t)} + \frac{a_t}{h(t)}\mu_t + \frac{a_t^2}{h(t)^2} C_t\, x = g_0(t) + g_1(t)\cdot x, \tag{47}$$

with $g_0(t) = a_t\mu_t/h(t)$ and $g_1(t) = -I_d/h(t) + a_t^2 C_t/h(t)^2$. Both g_0 and g_1 are real-analytic in t on $[t_{\max}, T]$ under standard noise schedules (linear or cosine β_t), so their Fourier expansions on $[t_{\max}, T]$ decay super-polynomially. Consequently, there exists $L_0 = L_0(\beta_t)$ such that for $L \ge L_0$,

$$\min_{U} \sup_{(x,t)} \left\| s_\theta^{\mathrm{HN}}(x,t) - s_{\mathrm{aff}}^*(x,t)\right\| \;\le\; O(L^{-r_{\mathrm{sched}}}), \tag{48}$$

where r_{sched} is the analytic decay rate of the Fourier coefficients. Combining with the cubic remainder from Lemma B.8, which is uniformly bounded by $R_{\mathcal{M}}^3$ on $\mathrm{supp}(p_t)$ since $a_t^3/h(t)^3 \le 1$ in Phase I, the total approximation error is absorbed into C_{HN}.

Step 2 (Estimation via Rademacher complexity). The hypothesis class is

$$\mathcal{F}_{\mathrm{HN}} := \left\{ s_\theta^{\mathrm{HN}} \ : \ \sum_\ell \|U_\ell\|_F^2 \le B^2 \right\}, \tag{49}$$

with B controlled by λ via the standard ridge bound $B \le \|s^*_{\mathrm{aff}}\|_{L^2}/\sqrt{\lambda}$. Since $\|\mu_t\| \le R_{\mathcal{M}}$ and $\|C_t\|_F \le \sqrt{d} \cdot R_{\mathcal{M}}^2$ (the latter using that C_t has d eigenvalues each bounded by $R_{\mathcal{M}}^2$),

$$\mathbb{E}_{t,x_t}\|s^*_{\mathrm{aff}}(x_t,t)\|^2 \ \le \ C \cdot d \cdot \mathrm{poly}(R_{\mathcal{M}}, h_{\min}^{-1}), \tag{50}$$

where $h_{\min} := h(t_{\max}) \asymp \tau^2$ is bounded away from zero throughout Phase I. By Lemma B.5 (adapted to the spatial input domain restricted to a ball of radius $O(\sqrt{d \cdot h_{\max}})$), the feature map has bounded ℓ^2 norm independently of d and m. The empirical Rademacher complexity is therefore

$$\widehat{\mathfrak{R}}_n(\mathcal{F}_{\mathrm{HN}}) \ \le \ \frac{B \cdot C_\sigma \sqrt{L}}{\sqrt{n}}. \tag{51}$$

Standard Rademacher-to-generalization conversion yields excess risk $O(B^2 C_\sigma^2 L \log(1/\delta)/n)$. Substituting $B^2 = O(d/\lambda)$ from the affine target's L^2 norm gives the parametric bound

$$\mathbb{E}\big\|s_{\hat\theta}^{\mathrm{HN}}(x_t,t) - s^*_{\mathrm{aff}}(x_t,t)\big\|^2 \ \le \ \frac{C_{\mathrm{HN}} \cdot d \cdot \log(1/\delta)}{n}. \tag{52}$$

Step 3 (Residual at T). Under the VP-SDE, the marginal p_T converges to $\mathcal{N}(0, I_d)$ exponentially fast. Specifically,

$$\mathrm{KL}(p_T \,\|\, \mathcal{N}(0, I_d)) \ \le \ C^{(2)}_{\mathcal{M}} \cdot e^{-T}, \tag{53}$$

which translates by Pinsker's inequality and Lemma B.8 into an $O(e^{-T})$ contribution in (46).

Combining Steps 1–3 yields the stated bound. □

End-to-end argument. Theorem 4.11 now follows by decomposing the reverse-SDE Girsanov bound across the two phases and applying the KL-to-W_2 interpolation inequality.

Proof of Theorem 4.11. We bound $\mathrm{KL}(p_{\mathrm{data},t_{\min}} \,\|\, p_{\mathrm{gen},t_{\min}})$ via Girsanov's theorem, then convert to W_2 through the standard interpolation inequality of [3].

Step 1 (Girsanov decomposition). Let $(\overleftarrow{x}_t)_{t\in[t_{\min},T]}$ denote the true reverse process and $(\overleftarrow{y}_t)$ the simulated process driven by $s_{\hat\theta}^{\mathrm{full}}$. Standard Girsanov estimates [15] yield

$$\mathrm{KL}\big(p_{\mathrm{data},t_{\min}} \,\|\, p_{\mathrm{gen},t_{\min}}\big) \ \le \ \frac{1}{2}\int_{t_{\min}}^{T} \mathbb{E}_{x_t\sim p_t}\big\|s_{\hat\theta}^{\mathrm{full}}(x_t,t) - s^*(x_t,t)\big\|^2 \, dt. \tag{54}$$

Step 2 (Phase decomposition). Using $\alpha(t) = \mathbf{1}[h(t) \le \tau^2]$, the gated architecture (15) coincides with $s_{\hat\theta}^{\mathrm{SiLD}}$ on $[t_{\min}, t_{\max}]$ and with $s_{\hat\theta}^{\mathrm{HN}}$ on $[t_{\max}, T]$. Splitting the integral in (54) at $t_{\max}$,

$$\begin{aligned}\mathrm{KL}\big(p_{\mathrm{data},t_{\min}} \,\|\, p_{\mathrm{gen},t_{\min}}\big) &\le \tfrac{1}{2}\int_{t_{\min}}^{t_{\max}} \mathbb{E}\big\|s_{\hat\theta}^{\mathrm{SiLD}} - s^*\big\|^2 dt + \tfrac{1}{2}\int_{t_{\max}}^{T} \mathbb{E}\big\|s_{\hat\theta}^{\mathrm{HN}} - s^*\big\|^2 dt \\ &\le \underbrace{O\!\left(\frac{\mathrm{poly}(k)\log(1/\delta)}{\sqrt{n}}\right)}_{\text{Phase I, by Theorems 4.5 and 4.8}} + \underbrace{O\!\left(\frac{C_{\mathrm{HN}} \cdot d \cdot \log(1/\delta)}{n} + e^{-T}\right)}_{\text{Phase II, by Lemma B.9}}. \end{aligned} \tag{55}$$

The Phase II term carries the e^{-T} damping because $p_t \to \mathcal{N}(0, I_d)$ exponentially as $t \to T$, so the score-error integrand on $[t_{\max}, T]$ is dominated by the contribution near T where the affine approximation is exact up to $O(e^{-T})$.

Step 3 (KL-to-W_2 conversion). With $t_{\min} \propto 1/n$, the interpolation inequality [3, Lemma 3] gives

$$W_2(p_{\mathrm{data}}, p_{\mathrm{gen},t_{\min}}) \ \lesssim \ \sqrt{\mathrm{KL}} \cdot n^{-1/4} + n^{-1/2}. \tag{56}$$

Substituting Step 2 yields (17), completing the proof. □

C Experimental Details and Additional Results

C.1 Toy experiment: full setup

We provide the complete specification of the synthetic experiment.

Data generation. We consider a manifold-plus-noise model in $\mathbb{R}^d$ with ambient dimension $d = 100$ and intrinsic dimension $k = 5$. Let $A \in \mathbb{R}^{d\times k}$ be a fixed orthonormal basis of the true manifold subspace, drawn uniformly from the Stiefel manifold once at the start of the experiment. Clean samples are generated as $x_0 = Az$, where the latent code $z \in \mathbb{R}^k$ follows a mixture of three Gaussians:

$$z \sim \sum_{c=1}^{3} \pi_c \mathcal{N}(\mu_c, 0.5^2 I_k),$$

with uniform mixing weights $\pi_c = 1/3$ and means $\{\mu_c\}_{c=1}^3$ placed at equidistant points on a circle of radius 2 in the first two coordinates of $\mathbb{R}^k$. Noisy observations are

$$x_t = x_0 + \sigma_t \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I_d),$$

with $\sigma_t = 0.1$, a small-noise regime in which the score singularity is pronounced and Proposition 3.1 applies.

Model. We train the two-stage score model of Section 3.3 with hidden width $m = 200$. Stage 1 uses the conservative-form network (2) with $\tanh$ activation. Stage 2 is a Random Feature network with frozen spatial features $V_x \in \mathbb{R}^{d\times m}$ drawn i.i.d. from $\mathcal{N}(0, I_d/d)$.

Optimization. Both stages are trained with Adam. Stage 1 uses learning rate $\eta_1 = 10^{-3}$; Stage 2 uses $\eta_2 = 5 \times 10^{-3}$. Batch size is 4096 for both stages. Stage 1 is trained until the orthogonal-component error plateaus, after which W is frozen and Stage 2 is trained on the ridge-regression objective of (13).

Diagnostics. Because the data-generating process is analytically tractable, we decompose the total score estimation error $\mathbb{E}\|\hat{s}(x_t) - s^*(x_t)\|_2^2$ into its manifold and orthogonal components exactly, using the ground-truth projection $\Pi_{\mathcal{M}} = AA^\top$. Writing $x_t = AA^\top x_t + (I - AA^\top)x_t$, the orthogonal component of the target score is $-(1/h_t)(I - AA^\top)x_t$, and the manifold component is the MoG score evaluated at $A^\top x_0$. The red and green curves in Figure 1 (left) track these two components independently throughout training, providing a direct empirical probe of the stage-by-stage learning dynamics predicted by Theorem 4.5.

C.2 Stacked MNIST qualitative samples

Figure 2 shows uncurated samples from both methods on Stacked MNIST, alongside real samples for reference. Both methods achieve full $1000/1000$ mode coverage (Table 1).

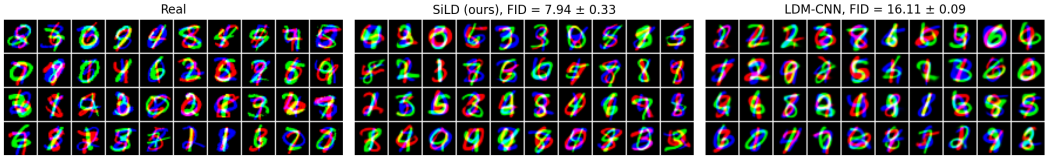


Figure 2: Uncurated samples from Stacked MNIST. Each image is three random MNIST digits stacked into the R/G/B channels (1000 possible modes). *Left:* real samples. *Middle:* SiLD. *Right:* LDM-CNN.

C.3 CelebA denoising comparison

Figure 3 shows the learned encoder-decoder applied to Gaussian-corrupted CelebA inputs without any diffusion steps, illustrating the Stage-1 objective in isolation. The SiLD Stage-1 loss trains the

encoder-decoder as a denoising operator on noisy inputs, while LDM's VAE objective optimizes reconstruction of clean inputs with KL regularization.



Figure 3: Denoising and reconstruction on CelebA (64×64). Encoder-decoder applied to Gaussian-corrupted inputs ($\sigma = 0.1$), no diffusion steps. *Row 1:* Original. *Row 2:* Noisy input. *Row 3:* LDM+GAN. *Row 4:* SiLD+GAN.

C.4 CelebA-HQ results

We extend the CelebA comparison to CelebA-HQ ($64 \times 64 \times 3$, ambient dimension $d = 12{,}288$, estimated intrinsic dimension $k_{95} = 223$ via PCA at 95% variance). All models use latent dimension 256. Table 4 reports results across two model sizes and two training budgets.

SiLD consistently outperforms LDM-CNN on reconstruction MSE across all settings, with the gap present at the smaller network (0.00440 vs. 0.00503) and persisting at the larger network (0.00345 vs. 0.00396). At $10\times$ training, both methods converge to near-identical reconstruction MSE (0.00061 vs. 0.00063), suggesting that reconstruction capacity saturates at sufficient optimization; at this scale LDM-CNN achieves slightly better Eps Loss (0.261 vs. 0.296). The overall pattern is consistent with CelebA: SiLD's Stage-1 objective produces a latent more faithful to the data manifold for reconstruction, while the generation metric depends additionally on the latent's amenability to diffusion.

Table 4: CelebA-HQ results ($64 \times 64 \times 3$, $d = 12{,}288$, intrinsic dim $k_{95} = 223$). All models use latent dim 256.

Model	Base Ch	AE Steps	Diff Steps	LPIPS	Eps Loss	Recon MSE
Small network (33M params)						
SiLD (ours)	64	10k	30k	0.5	0.380	**0.00440**
LDM-CNN	64	10k	30k	0.5	**0.305**	0.00503
Larger network (104M params)						
SiLD (ours)	128	10k	30k	0.5	0.326	**0.00345**
LDM-CNN	128	10k	30k	0.5	**0.271**	0.00396
Larger network + 10× training						
SiLD (ours)	128	100k	300k	0.5	0.296	**0.00061**
LDM-CNN	128	100k	300k	0.5	**0.261**	0.00063

C.5 Compute Resources

All experiments were conducted on a single NVIDIA A100 (40 GB) GPU; times below are wall-clock training time per configuration, excluding scheduler queue waits. Stacked MNIST (Table 1)

takes ≈ 50 min. CelebA (64×64, Table 2) ranges from 1–2 h for the LDM/SiLD baselines, ≈ 3.3 h for the MMD α-sweep, and ≈ 12.4 h for each GAN-regularized row, totalling ≈ 22 GPU-hours. MoleculeNet (Table 3, four datasets) takes 0.4–2.6 h per (method, dataset) configuration, totalling ≈ 16.5 GPU-hours. CelebA-HQ (Table 4) base configurations run in 1–2 h; the $10\times$ training row (100k AE + 300k diffusion steps, 104M parameters) is the most expensive single configuration at ≈ 17 h per method, with the table totalling ≈ 49 GPU-hours. The reported experiments thus account for ≈ 90 GPU-hours; including ablations, hyperparameter sweeps, and preliminary architecture exploration, total measured wall time across all jobs is approximately 320 GPU-hours.